

BCA Science (Third Year) (w.e.f.2026-27)
Semester-V

S. No	Course Code	Name of the Course	Course Type	Weekly Workload, Hrs.				Credits	Assessment Scheme Code
				L	T	P	J		
1	BCA30030	Introduction to Machine Learning Using Python	PM	3		2		4	TL3
2	BCA20110	Programming in JAVA	PM	3		2		4	TL3
3	BCA30040	Introduction to Cloud Computing	PM	2				2	TT1
4		Program Elective-I	PE	3	1			4	TT1
5	BCA20120	Software Engineering	PM	3				3	TT1
6	PCE10030	Managing Conflicts Peacefully: Tools and Techniques	UC	2				2	TT1
7	BCA30050	Project Based Learning - III	PR				3	1	PJ
		Total		16	1	4	3	20	

MIT-WPU, PUNE-38.

APPROVED BY

05 JUN 2024

ACADEMIC COUNCIL

Weekly Teaching Hours: 24

Total Credits: 20

** Assessment Marks are valid only if Attendance criteria are met
L-Lecture, T-Tutorial, P-Practical, J-Project.



Academic
Coordinator



Program Director/
HOS



Associate Dean
Academics



Dean



Dean Academics
MITWPU



Registrar
MITWPU

BCA Science (Third Year) (w.e.f.2026-27)
Semester-VI

S. No	Course Code	Name of the Course	Course Type	Weekly Workload, Hrs.				Credits	Assessment Scheme Code
				L	T	P	J		
1	BCA30060	Internet Of Things	PM	2		2		3	TL4
2	BCA10070	Operational Research	PM	3				3	TT1
3	BCA30070	Advanced Java	PM	3		2		4	TL3
4	BCA30080	Introduction to Blockchain Technology	PM	3				3	TT1
5		Program Elective -II	PE	3	1			4	TT1
6	BCA40010	Project Based Learning - IV	PR				3	1	PJ
7	UNC10140	National Academic Immersion	UC	2				2	I
Total:				16	1	4	3	20	

MIT-WPU, PUNE-38
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05 JUN 2025

ACADEMIC COUNCIL

** Assessment Marks are valid only if Attendance criteria are met
L-Lecture, T-Tutorial, P-Practical, J-Project.

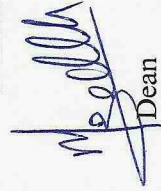
Weekly Teaching Hours: 24

Total Credits: 20


Academic
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MITWPU

Syllabus

**Dr. Vishwanath Karad
MIT- World Peace University**

**FACULTY OF ENGINEERING AND
TECHNOLOGY**

**Bachelor of Computer Applications
BCA (Semester-V)**



ep B. Karad

COURSE STRUCTURE

CourseCode	BCA30030				
CourseCategory	Program Major				
CourseTitle	Introduction to Machine Learning using Python				
Teaching Scheme	Lecture s	Tutorial s	Laboratory / Practical	Projec t	Total
Weekly load hours	3	-	2	-	5
Credits	3	-	1	-	4
Assessment Schema Code	TL3				

Pre-requisites:

Course Objectives:

10. To develop a comprehensive understanding of the principles machine learning
11. To develop the skills to apply essential data preprocessing and EDA techniques
12. To cultivate the ability to choose suitable machine learning technique and algorithm for real-world data analysis problems.
13. To introduce machine learning model performance metrics

Course Outcomes:

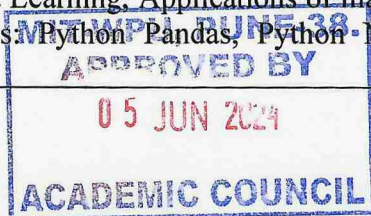
On completion of the course, student will be able to–

1. Understand the stages of machine learning project
2. Choose and apply appropriate data preprocessing and EDA techniques
3. Select and apply appropriate supervised machine learning technique and algorithm for real day problems
4. Evaluate and compare the performance of various machine learning algorithms

Course Contents:

Unit 1: Introduction to Machine Learning & Python [9]

Traditional Programming Vs Machine Learning, Steps of Machine Learning Process, Types of machine learning : Supervised Machine Learning , Unsupervised Machine Learning ,Semi-Supervised Machine Learning, and Reinforcement Learning. Applications of machine learning, Python for Data Science, Essential libraries and tools Python Pandas, Python Numpy, Python SciPy, Python Matplotlib



Unit 2: Data Preprocessing and EDA [9]

Data set : attribute types, explanatory variable, outcome variable, Data Preprocessing : Handling missing data, smooth out noise, handling outliers, encoding categorical variables, feature scaling, normalization, discretization, Why EDA? Univariate graphical EDA : Histograms, Boxplots Univariate non-graphical EDA : Frequency distribution table, Measures, of central tendency, measures dispersion, Multivariate non-graphical EDA: cross-tabulation, correlation matrix, Multivariate graphical EDA: Scatter plot, Heat Map, Why Feature Selection, Feature selection techniques

Unit 3: Supervised machine Learning : Regression [9]

Regression, independent variable, dependent variable, Simple Linear Regression, Multiple Linear Regression, Polynomial Regression, Ridge Regression, Lasso Regression, Support Vector Regression, Decision Tree Regression, Random Forest Regression, Evaluating Regression models : Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), R-squared, adjusted R- square

Unit 4: Supervised machine Learning : Classification [9]

Classification process, classification Vs Regression, Logistic Regression, Decision Tree Classification, Random Forest Classification, Support Vector Classification, Confusion matrix, Evaluating Classification models : Accuracy, Precision, Recall, F1 score

Unit 5: Unsupervised machine Learning : [9]

Clustering, Cluster analysis, Overview of clustering methods, k-means, Hierarchical clustering, Accuracy measures : Silhouette Score, Calinski-Harabasz Index, Inertia

Learning Resources:

Textbooks/Reference Books:

1. Introduction to Machine Learning, Ethem Alpaydin, 3rd Edition, MIT Press, 2014
2. An Introduction to Machine Learning, Miroslav Kuba, 3rd Edition, Springer International Publishing, 2021

Supplementary Reading:

1. Pattern Recognition and Machine Learning, C. M. Bishop, 1st Edition, Springer, 2013
2. Introduction to Machine Learning with Python, Andreas C. Müller, Sarah Guido, 3rd Edition, O'Reilly Media, 2017

Weblinks:

1. <https://www.geeksforgeeks.org/introduction-machine-learning/>
2. <https://www.javatpoint.com/machine-learning>

MOOCs: Online courses for self-learning

Courses by NPTEL and MIT Open Courseware etc

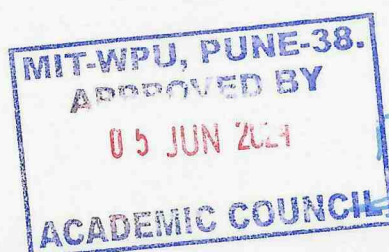


Pedagogy:

1. Participative Learning,
2. discussions,
3. algorithm,
4. programming concepts,
5. experiential learning through practical problem solving,
6. assignments.

Laboratory Experiments / Software based Practical

Sr No.	Contents
1	Python Data Science Environment Setup, Essential libraries and tools: Python Pandas, Python Numpy, Python SciPy, Python Matplotlib
2	Implementation of EDA
3	Implementation of Supervised machine Learning : Regression
4	Implementation of Supervised machine Learning : Classification
5	Implementation of Unsupervised machine Learning



COURSE STRUCTURE

Course Code	BCA20110				
Course Category	Program Major				
Course Title	Programming in JAVA				
Teaching Scheme and Credits	L	T	Laboratory/Practical	Project	Total
Weekly load hrs.	3	-	2	-	5
Credits	3	-	1	-	4
Assessment Schema Code	TL3				
<u>Pre-requisites:</u> Need basics of Programming and Object-Oriented Programming concepts.					
Course Objectives: Knowledge: To understand Java programming concepts with respect to i) Object Oriented Programming ii) Data Structures in Java Programming Skills: Design skills of i) Desktop Applications Development ii) Object oriented programming Attitude: To develop following i) Programming Skills ii) Build Stand-alone Application in Complex situation					
<u>Course Outcomes:</u> a) Identify classes, objects, class members and relationships for a given problem. b) Design end to end applications using object-oriented constructs. c) Apply collection classes for storing java objects. d) Use Java APIs for program development. e) Handle abnormal termination of a program using exception handling.					
Course Contents Unit 1: Introducing to Java:4 A Short History of Java, Features of Java, Java Environment – Compiler, Interpreter, JVM, Simple java, program, Types of Comments, Declaring single and multi-dimensional arrays, Accepting input using Command line arguments, Accepting input from console (Using BufferedReader and Scanner class) Unit 2: Classes and Objects:8 JSP Basics, JSP Derivatives, Implicit Object in JSP, JSP Actions, JSP Database Handling, Session, Handling in JSP, JSP Beans, JSP Custom Tag Packages -Creating, Accessing and using Packages Unit 3: Inheritance:10 Inheritance Basics (extends Keyword) and Types of Inheritance, Superclass, Subclass and use of Super Keyword, Method Overriding and runtime polymorphism. Use of final keyword related to					

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05 JUN 2021

variable, method and class, Use of abstract class and abstract methods

Interface and Packages-Defining and Implementing Interfaces, Runtime polymorphism using interface

Unit 4: Collections -8

Wrapper Classes, Introduction to the Collection framework, List – ArrayList, LinkedList and Vector,

Set - HashSet, TreeSet, and LinkedHashSet, Map – HashMap, LinkedHashMap, Hashtable and TreeMap

Interfaces such as Iterators, ListIterators, Enumerations

Unit 5: Exception Handling :4

Exception class, Checked and Unchecked exception, Catching exception and exception handling – try, catch, finally, throw and throws, multiple catch block, Creating user defined exception

Unit 6: I/O:3

String class(basic methods) , String Buffer class File class DataInputStream and DataOutputStream class

Unit 7:Swing:8

What is Swing? The MVC Architecture and Swing, Layout Manager and Layouts, The JComponent class Components – JLabel, JButton, JText, JTextArea, JCheckBox, JRadioButton, JList, JComboBox, JMenu and JPopupMenu Class, JMenuItem, Dialogs (Message, confirmation, input), JFileChooser, Event Handling: Event sources, Listeners – ActionListener, ItemListener, Mouse and Keyboard Event Handling, Adapters – MouseAdapter, KeyAdapter, Anonymous inner class

Learning Resources:

Reference Books:

1. Java 2 programming black books, Steven Horlizer

Supplementary Reading:

2. Complete reference Java by Herbert Schildt(5th edition)

3. Java Complete Reference Patric Naughton, Herbert Schildt, TMH,7th Ed.

4. Complete Reference- J2EE Jim Keogh, TMH.

5. Inside Servlets- Dustine R Callway

7. Programming with Java, A primer, Fourth edition, By E. Balagurusamy

8. Core Java Volume-I-Fundamentals, Eighth Edition, Cay S. Horstmann, Gary Cornell, Additional Reading

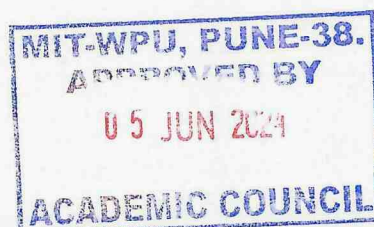
Websites : <https://www.tutorialspoint.com>

<https://www.javatpoint.com>

MOOCS : Spoken Tutorial, Khan Academy

Pedagogy:

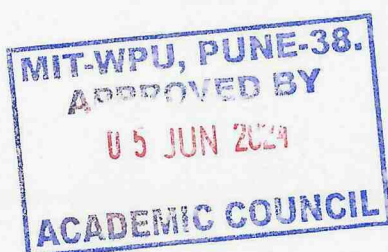
- Practical learning
- project handling
- problem solving assignment
- conceptual and contextual learning
- PPT Presentation & Interactive discussion



Laboratory Experiments / Software based Practical

Sr No.	Contents
1	Implementation of basic concepts of Java, constants, operators, decisions and loops etc.
2	Implementation of Classes, Objects, Creation of objects in various ways, constructors, Packages
3	Implementation of Inheritance, types of inheritance, aggregations, Interface, Polymorphism and Packages
4	To study Collection framework.
5	To study Exception, types of exceptions, throw, throws.
6	I/O Handling-String and StringBuffer class, File class, DataInputStream and DataOutputStream class
7	To Study GUI development using SWING

M. Bechu



AP. Bechu

COURSE STRUCTURE

Course Code	BCA30040				
Course Category	Program Major				
Course Title	Introduction to Cloud Computing				
Teaching Scheme	Lectures	Tutorial	Laboratory / Practical	Project	Total
Weekly load hours	2	-	-	-	2
Credits	2	-	-	-	2
Assessment Scheme Code	TT1				

Pr-requisites:

1. Basic knowledge of computers

Course Objectives:

1. To understand the concept of Cloud Computing.
2. To be familiar with components required for different types of Cloud Services.
3. To be familiar with different Cloud Deployment Model.

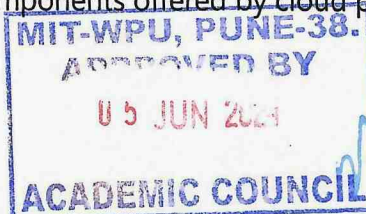
Course Outcomes:

Student will able to

1. Students will gain a comprehensive understanding of what cloud computing is, including its definition, characteristics, and essential components.
2. Students will learn about different cloud service models such as Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS), including their characteristics, advantages, and use cases.
3. Students will be introduced to various cloud deployment models, including public cloud, private cloud, hybrid cloud, and community cloud, and understand when each deployment model is appropriate.
4. Students will develop skills in designing cloud architectures, including understanding the principles of scalability, elasticity, reliability, and security in cloud-based systems.
5. Students will become familiar with major cloud service providers such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP), including their offerings, pricing models, and basic usage.
6. Students will gain practical experience by working with various cloud services such as virtual machines, storage solutions, databases, and networking components offered by cloud providers.

Course Contents:

Unit 1 : Understanding Cloud Computing: 6
Introduction to Cloud Computing



Characteristics of Cloud Computing
Evolution of Cloud Computing
Cloud Computing Use Cases
Benefits and Challenges of Cloud Computing

Unit 2 : Cloud Service Models: 6

Infrastructure as a Service (IaaS)
Platform as a Service (PaaS)
Software as a Service (SaaS)
Function as a Service (FaaS)
Comparison of Cloud Service Models

Unit 3 : Cloud Deployment Models: 6

Public Cloud
Private Cloud
Hybrid Cloud
Community Cloud
Choosing the Right Deployment Model

Unit 4 : Major Cloud Providers: 6

Amazon Web Services (AWS)
Microsoft Azure
Google Cloud Platform (GCP)
IBM Cloud
Oracle Cloud

Unit 5: Cloud Storage and Networking: 6

Cloud Storage Services
Content Delivery Networks (CDNs)
Virtual Private Clouds (VPCs)
Cloud Networking Concepts
Advanced Networking in the Cloud

Learning Resources:

Text Books/Reference Books:

- "Cloud Computing: Concepts, Technology & Architecture" by Thomas Erl, Ricardo Puttini, and Zaigham Mahmood
- "Cloud Computing For Dummies" by Judith Hurwitz, Robin Bloor, Marcia Kaufman, and Fern Halper
- "Architecting the Cloud: Design Decisions for Cloud Computing Service Models (SaaS, PaaS, and IaaS)" by Michael J. Kavis

Supplementary Reading:

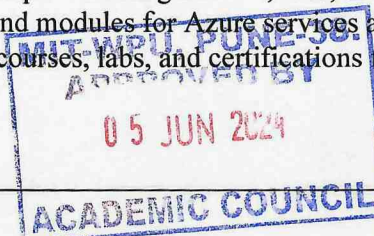
- AWS Documentation: Comprehensive documentation

Web Resources :

Weblinks:

- AWS Training and Certification: Offers free and paid training courses, labs, and certification exams for AWS.
- Microsoft Learn: Provides free learning paths and modules for Azure services and concepts.
- Google Cloud Training: Offers online training courses, labs, and certifications for Google Cloud Platform (GCP).

MOOCs: Online courses for self-learning
Courses by NPTEL and MIT Open



Courseware etc Pedagogy:

- Participative Learning,
- discussions,
- problem solving,
- assignments,
- Lab Assignment

M. Beck

EP 100

COURSE STRUCTURE

Course Code	BCA20120				
Course Category	Program Major				
Course Title	Software Engineering				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3	-	-	-	3
Credits	3	-	-	-	3
Assessment Scheme Code	TT1				

Pre-requisites:

Understanding of Computer Science Fundamentals, Software development

Course Objectives:

1. To understand the knowledge and skills necessary for effective software development and management.
2. To Understanding Software Engineering
3. To develop the ability to gather, analyze, and document software requirements

Course Outcomes:

1. To articulate the specific knowledge, skills, and abilities that students are expected to gain by the end of the course.
2. Requirements Engineering:
3. Elicit, analyze, and document software requirements effectively, considering both functional and non-functional aspects.
4. System Design Proficiency:
5. Design software systems using appropriate architectural styles, patterns, and object-oriented design principles.

Course Contents:

Unit 1: Introduction to Software Engineering[2]

Definition and goals of software engineering, Software development life cycle models, Roles and responsibilities in software development teams, Ethical and professional issues in software engineering

Unit 2: Software Processes: [3]

Software process models , Process activities, Coping with change, Process improvement

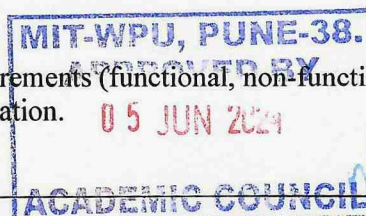
Unit 3: Agile Methodologies[5]

Agile methods, Agile development techniques, Agile project management, Scaling agile methods

Unit 4: Software Requirements[5]

Requirement elicitation and analysis, Types of requirements (functional, non-functional) Use cases and scenarios, Requirement specification, and documentation.

Unit 5: System Design[5]



Architectural design, Component-level design, Object-oriented design principles, Design patterns
Design documentation

Unit 6: Implementation and Coding[3]

Programming languages and paradigms, Code organization and structure, Coding standards and best practices, Code reviews and inspections.

Unit 5: Software Testing[3]

Testing levels (unit, integration, system, acceptance), Testing types (functional, non-functional), Test planning and strategy, Test automation.

Unit 6: Software Maintenance and Evolution[4]

Software maintenance and its importance, Software reengineering and reverse engineering, Version control systems, and Code refactoring.

Unit 7: Software Project Management:[5]

Project planning and scheduling, Risk management, Team collaboration and communication
Agile and Scrum methodologies.

Unit 8: Software Quality Assurance[5]

Quality metrics and measurement, Continuous integration and continuous delivery (CI/CD), DevOps practices.

Unit 9: Emerging Trends in Software Engineering[5]

Cloud computing and its impact on software development, Artificial intelligence in software engineering, Internet of Things (IoT) and software engineering

Learning Resources:

Text Books/Reference Books:

1. SOFTWARE ENGINEERING: A PRACTITIONER'S APPROACH, 7TH EDITION"
2. Software Engineering: A PRACTITIONER'S APPROACH, By Rajiv Chopra
3. GitHub Learning Lab-Provides hands-on, project-based learning using GitHub.
4. LeetCode-Platform for practicing coding problems and preparing for technical interviews.

Web Resources :

Weblinks:

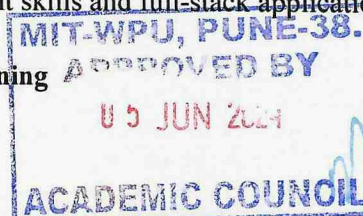
1. Coursera - Software Engineering Specialization (offered by University of British Columbia)
 - A comprehensive specialization covering various aspects of software engineering.
2. edX - Software Development and Testing MicroMasters Program (offered by the University of British Columbia)
 - A series of courses covering software development, testing, and best practices.
3. Udacity - Full Stack Web Developer Nanodegree
 - Focuses on web development skills and full-stack applications.

MOOCs: Online courses for self-learning

Courses by NPTEL and MIT
Open Courseware etc

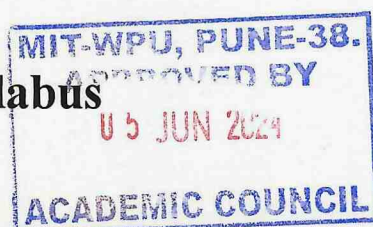
Pedagogy:

- Participative Learning,



- Discussions,
- Problem solving,
- Assignments.

Syllabus



M. Bechu
EP. Kholi

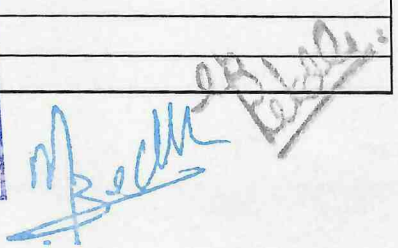
Dr. Vishwanath Karad
MIT- World Peace University

**FACULTY OF ENGINEERING AND
TECHNOLOGY**

Bachelor of Computer Applications
BCA (Semester-VI)

BATCH: 2024-28

COURSE
STRUCTURE

Course Code	BCA30060	MIT-WPU, PUNE-38. APPROVED BY 05 JUN 2024 ACADEMIC COUNCIL 
Course Category	Program Major	
Course Title	Internet of Things	

Teaching Scheme	Lectures	Tutorial	Laboratory / Practical	Project	Total
Weekly load hours	2	-	2	-	4
Credits	2	-	1	-	3
Assessment Scheme Code	TL4				

Pr-requisites:

1. Basic knowledge of computers

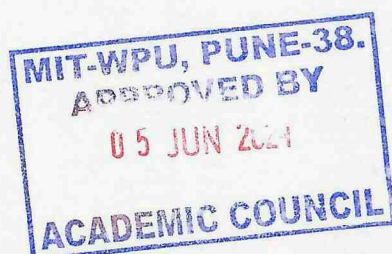
Course Objectives:

4. To understand the IoT Fundamentals:
5. To Develop the Practical IoT Skills:
6. To Address the IoT Challenges and Opportunities:

Course Outcomes:

Student will able to

7. Students will Gain a foundational understanding of IoT concepts, including its definition, components, and architecture.
8. Students will be Familiar with IoT hardware components such as sensors, actuators, microcontrollers, and communication modules.
9. Students will Understand IoT communication protocols like MQTT, CoAP, and HTTP, as well as wireless standards such as Zigbee, Bluetooth, and LoRa..
10. Students will be Abile to collect, process, and analyze data generated by IoT devices using platforms like Arduino, Raspberry Pi, or other microcontroller platforms.



M. Beek
SP. Beek

Course Contents:

Unit 1 : Introduction to IoT: 3

Definition and scope of IoT
Evolution and history of IoT
Applications and use cases

Unit 2 : IoT Architecture and Components: 5

IoT hardware components: sensors, actuators, microcontrollers
IoT software components: operating systems, middleware, applications
IoT communication protocols: MQTT, CoAP, HTTP, etc.

Unit 3 : Data Acquisition and Processing: 4

Data collection techniques
Data preprocessing and filtering
Data storage and management

Unit 4 : IoT Security: 4

Security threats and vulnerabilities in IoT systems
Authentication and authorization mechanisms
Encryption and data integrity

Unit 5: Cloud Integration: 5

Cloud computing basics
IoT and cloud platform integration
Data storage, processing, and analysis in the cloud

Unit 6: Edge Computing: 5

Edge computing fundamentals
Benefits and challenges of edge computing in IoT
Edge computing architectures and frameworks

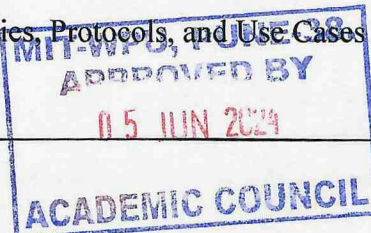
Unit 7: IoT Applications: 4

Healthcare IoT
Smart cities and urban IoT
Industrial IoT (IIoT)
Agriculture and environmental monitoring

Learning Resources:

Text Books/Reference Books:

1. "Building the Internet of Things: Implement New Business Models, Disrupt Competitors, Transform Your Industry" by Maciej Kranz
2. "Internet of Things (IoT): Technologies, Applications, Challenges and Solutions" edited by B.B. Gupta and Dharma P. Agrawal
3. "IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things" by David Hanes and Gonzalo Salgueiro



Supplementary Reading:

Practical Internet of Things Security" by Brian Russell and Drew Van Duren

Web Resources :

Weblinks:

1. **IoT For All:** A comprehensive online platform offering articles, guides, webinars, and case studies covering various aspects of IoT, including technology, applications, trends, and industry insights. Website: [IoT For All](#)
2. **IEEE Internet of Things (IoT) Initiative:** The IEEE IoT Initiative provides a wealth of resources including articles, publications, conferences, and standards related to IoT technology and applications. Website: [IEEE IoT Initiative](#)
3. **IoT Agenda:** A dedicated section on TechTarget's website covering news, articles, expert advice, and opinion pieces on IoT technology, trends, and best practices. Website: [IoT Agenda](#)
4. **IoT World Today:** An online publication focusing on IoT news, insights, and analysis across various industries such as healthcare, manufacturing, smart cities, and more. Website: [IoT World Today](#)

MOOCs: Online courses for self-learning

Courses by NPTEL and MIT Open

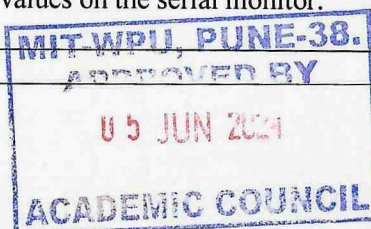
Courseware etc

Pedagogy:

- Participative Learning,
- discussions,
- problem solving,
- assignments,
- Lab Assignment

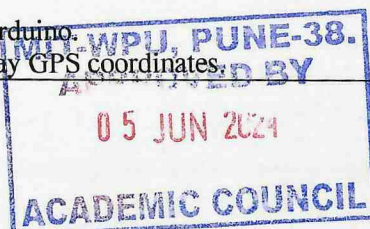
Laboratory Experiments / Software based Practical:

Sr.No	Experiments to be conducted on
1	Introduction to Arduino and Basic LED Control Objective: Learn the basics of Arduino programming and control an LED. Materials: Arduino Uno, LED, resistor, breadboard, jumper wires. Procedure: • Set up Arduino with an LED. • Write a simple program to blink the LED. • Modify the program to adjust the blinking rate.
2	Temperature Sensing with DHT11 Objective: Interface a DHT11 sensor to measure temperature and humidity. Materials: Arduino Uno, DHT11 sensor, breadboard, jumper wires. Procedure: • Connect the DHT11 sensor to the Arduino. • Write a program to read data from the sensor. • Display temperature and humidity values on the serial monitor.
3	. Light Sensing with LDR



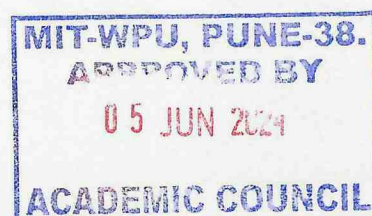
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	Objective: Use an LDR (Light Dependent Resistor) to measure light intensity. Materials: Arduino Uno, LDR, resistor, breadboard, jumper wires. Procedure: • Connect the LDR to the Arduino. • Write a program to read the light intensity. • Display the light intensity values on the serial monitor.
4	Motion Detection with PIR Sensor Objective: Detect motion using a PIR sensor. Materials: Arduino Uno, PIR sensor, breadboard, jumper wires. Procedure: • Connect the PIR sensor to the Arduino. • Write a program to detect motion. • Turn on an LED when motion is detected.
5	Remote LED Control via Bluetooth Objective: Control an LED using a smartphone via Bluetooth. Materials: Arduino Uno, Bluetooth module (HC-05), LED, resistor, breadboard, jumper wires. Procedure: • Connect the Bluetooth module and LED to the Arduino. • Pair the Bluetooth module with a smartphone. • Use a mobile app to send commands to control the LED.
6	Sending Data to ThingSpeak Objective: Send sensor data to the ThingSpeak cloud platform. Materials: Arduino Uno, ESP8266 Wi-Fi module, DHT11 sensor. Procedure: • Connect the ESP8266 and DHT11 to the Arduino. • Write a program to read sensor data and send it to ThingSpeak. • Visualize the data on ThingSpeak dashboards.
7	Basic Web Server with ESP8266 Objective: Create a simple web server to control an LED. Materials: ESP8266 module, LED, resistor, breadboard, jumper wires. Procedure: • Set up the ESP8266 as a web server. • Write a program to control the LED via a web interface. • Test the web server on a local network.
8	Home Automation with MQTT Objective: Use MQTT protocol for home automation. Materials: Arduino Uno, ESP8266 module, relay module, MQTT broker (e.g., Mosquitto). Procedure: • Connect the ESP8266 and relay to the Arduino. • Set up an MQTT broker. • Write a program to control the relay using MQTT messages.
9	Soil Moisture Monitoring Objective: Measure soil moisture using a soil moisture sensor. Materials: Arduino Uno, soil moisture sensor, breadboard, jumper wires. Procedure: • Connect the soil moisture sensor to the Arduino. • Write a program to read and display soil moisture levels. • Add an LED to indicate low moisture levels.
10	GPS Tracking with Arduino Objective: Track location using a GPS module. Materials: Arduino Uno, GPS module (e.g., Neo-6M), breadboard, jumper wires. Procedure: • Connect the GPS module to the Arduino. • Write a program to read and display GPS coordinates.



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	• Display the coordinates on the serial monitor.
11	Air Quality Monitoring Objective: Monitor air quality using an MQ-135 sensor. Materials: Arduino Uno, MQ-135 air quality sensor, breadboard, jumper wires. Procedure: • Connect the MQ-135 sensor to the Arduino. • Write a program to read and display air quality levels. • Indicate poor air quality with an LED.
12	Smart Door Lock with RFID Objective: Implement a smart door lock using RFID. Materials: Arduino Uno, RFID module (RC522), servo motor, breadboard, jumper wires. Procedure: • Connect the RFID module and servo motor to the Arduino. • Write a program to unlock the door with a valid RFID tag. • Test the system with different RFID tags.
13	Heart Rate Monitoring with Pulse Sensor Objective: Measure heart rate using a pulse sensor. Materials: Arduino Uno, pulse sensor, breadboard, jumper wires. Procedure: • Connect the pulse sensor to the Arduino. • Write a program to read and display heart rate. • Visualize the heart rate data on the serial monitor.
14	Smart Irrigation System Objective: Automate plant watering based on soil moisture. Materials: Arduino Uno, soil moisture sensor, relay module, water pump, breadboard, jumper wires. Procedure: • Connect the soil moisture sensor and relay to the Arduino. • Write a program to activate the water pump when soil moisture is low. Test the system in a plant pot.
15	Voice-Controlled Home Automation Objective: Control home appliances using voice commands. Materials: Arduino Uno, Bluetooth module, relay module, microphone module, breadboard, jumper wires. Procedure: • Connect the Bluetooth and relay modules to the Arduino. • Pair the Bluetooth module with a voice command app. • Write a program to control the relay based on voice commands.



COURSE STRUCTURE

Course Code	BCA10070				
Course Category	Program Major				
Course Title	Operational Research				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3	-	-	-	3
Credits	3	-	-	-	3
Assessment Scheme Code	TT1				

Pre-requisites:

Understanding of Computer Science Fundamentals, Software development

Course Objectives:

1. CO1 Identify and develop operations research model describing a real-life problem.
2. CO2 Understand the mathematical tools that are needed to solve various optimization problems.
3. CO3 Solve various linear programming, transportation, assignment, queuing, inventory and game problems related to real life.

Course Outcomes:

1. To know optimizing techniques
2. To understand its use in decision making in business
3. To Identify and develop operational research model from real system
4. To appreciate the mathematical basis for business decision making

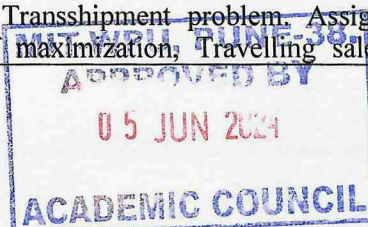
Course Contents:

Unit 1: Operations Research: [10]

Origin, Definition and scope. Linear Programming: Formulation and solution of linear programming problems by graphical and simplex methods, Big - M and two-phase methods, Degeneracy, Duality in linear programming

Unit 2: Transportation Problems: [13]

Basic feasible solutions, Optimum solution by stepping stone and modified distribution methods, Unbalanced and degenerate problems, Transshipment problem, Assignment problems: Hungarian method, Unbalanced problem, Case of maximization, Travelling salesman and crew assignment



problems.

Unit 3: [10]

Concepts of stochastic processes, Poisson process, Birth-death process, Queuing models: Basic components of a queuing system, Steady-state solution of Markovian queuing models with single and multiple servers (M/M/1, M/M/C, M/M/1/k, M/MC/k)

Unit 4: Inventory control models:[12]

Economic order quantity (EOQ) model with uniform demand, EOQ when shortages are allowed, EOQ with uniform replenishment, Inventory control with price breaks. Game Theory: Two-person zero sum game, Game with saddle points, the rule of dominance; Algebraic, Graphical and linear programming methods for solving mixed strategy games.

Learning resources:

Text Books/Reference Books:

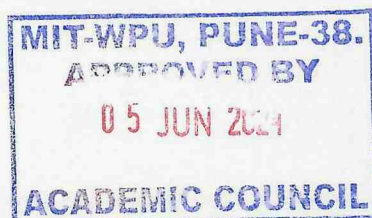
1. H.A. Taha, Operation Research-An introduction, Printice Hall of India.
2. P.K. Gupta and D.S. Hira, Operations Research, S. Chand & Co.
3. S.D. Sharma, Operation Research, Kedar Nath Ram Nath Publications.
4. J.K. Sharma, Mathematical Model in Operation Research, Tata McGraw Hill.

Web Resources :

1. <https://www.classcentral.com/course/swayam-operations-research-14219>
2. https://onlinecourses.nptel.ac.in/noc19_ma29/preview

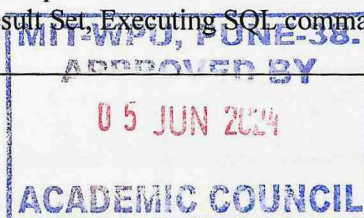
Pedagogy:

- Participative Learning,
- discussions,
- problem solving,
- assignments,
- Lab Assignment



COURSE STRUCTURE

Course Code	BCA30070			
Course Category	Program Major			
Course Title	Advanced Java			
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Total
Weekly load hrs.	3	-	2	5
Credits	3		1	4
Assessment Scheme code	TL3			
<u>Pre-requisites:</u> Need basics Java Programming, Database and Web Technology concepts				
Course Objectives: Knowledge: To understand java programming concepts with respect to i) Server-Side Programming ii) Web Development Skills: Design skills of i) Web Base Application ii) Distributed Application Attitude: To develop following i) Programming Skills ii) Build Web Application in Complex situation				
<u>Course Outcomes:</u> 1. Understand the concepts related to Java Technology. 2. Explore and understand use of Java Server Programming. 3. Create dynamic web pages, using Servlets and JSP, Web Services, Spring 4. Framework and Hibernate Framework. 5. Students learn skills to develop real time applications 6. Students learn to access database through Java programs, using Java Database Connectivity (JDBC) 7. Develop advanced skills for programming in Java				
Course Contents: Unit 1: JDBC:6 Introduction to JDBC, Basic JDBC program Concept Drivers, Making the Connection, Statement, Result Set, Executing SQL commands, Executing queries.				



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Unit 2:Multithreading:3

Threading basics, Life cycle of thread,Creating Threads, Priorities and Synchronization, Inter Thread Communication Runnable Interface.

Unit 3:Basic Concepts of Servlet:8

Servlet Life Cycle, Coding and writing servlet, programme, Filter, Request Dispatcher, Session Tracking in Servlet, Servlet and JDBC, Writing ThreadSafe Servlet

Unit 4:JSP:8

JSP Basics, JSP Derivatives, Implicit Object in JSP, JSP Actions, JSP Database Handling, Session Handling in JSP, JSP Custom Tag

Unit 5:Web Services:5

Web services concepts, WSDL, UDDI, Introduction to SOAP, XML-RPC, Creating web services, Calling web services

Unit6: Spring:8

Spring Framework, Inversion of Control, Dependency Control, Aspect Oriented Programming, Bean Factory, Error Handling

Unit7:Hibernate:7

Introduction, Hibernate with Spring, Hibernate Query Language, ORA / Hibernate Mapping, Hibernate in Web Application

Learning Resources:**Reference Books:**

1. Java 2 programming black books, Steven Horlzner

Supplementary Reading:

1. Complete reference Java by Herbert Schildt(5th edition)
2. Java Complete Reference Patric Naughton, Herbert Schildt, TMH,7th Ed.
3. Complete Reference- J2EE Jim Keogh, TMH.
4. Inside Servlets- Dustine R Callway
5. Spring and Hibernate, Santosh Kumar K. Mc.Graw Hill Education
6. Programming with Java, A primer, Fourth edition, By E. Balagurusamy
7. Core Java Volume-I-Fundamentals, Eighth Edition, Cay S. Horstmann, Gary Cornell,

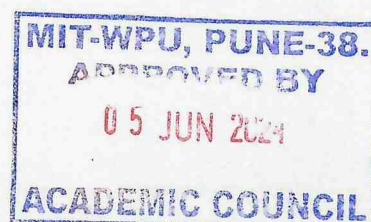
Additional Reading

Websites : <https://www.tutorialspoint.com> , www.javatpoint.com

MOOCS : Spoken Tutorial, Khan Academy

Pedagogy:

- Practical learning,
- project handling,
- problem solving assignment,
- conceptual and contextual learning,
- PPT Presentation & Interactive discussion



Laboratory Experiments / Software based Practical

Sr No.	Contents
1	To Make a Connections with Database using JDBC, execute DML Commands using Statement and Prepared Statement Interfaces, Store the result into ResultSet. Display and Traverse the records using collection objects
2	To implement creation of thread in different along with collection
3	Implementation of To study Servlet creation, Servlet and JDBC
4	To implement JSP for creating web pages with JDBC
5	To implement user microservice, rating microservice, implementing service registry, microservices communications etc
6	To implement spring concepts, IOC, DI, DI with collection, autowiring, AOP
7	To implement hibernate for ORM, HB with annotation, HB mapping with collection

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05 JUN 2024
ACADEMIC COUNCIL

M. Beck
EB Blue

COURSE STRUCTURE

Course Code	BCA30080				
Course Category	Program Major				
Course Title	Introduction to Blockchain Technology				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3	-	-	-	3
Credits	3	-	-	-	3
Assessment Schema Code	TT1				

Pre-requisites:

Basic knowledge of mathematical and algorithmic logic

Course Objectives:

1. To understand the concepts of Blockchain Ecosystem.
2. To understand how to use blockchain technology in day-to-day problem-solving.
3. To develop knowledge of Blockchain fundamentals.

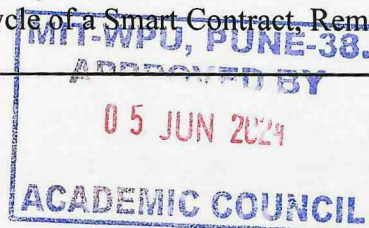
Course Outcomes:

After completion of this course, students will be able to:

1. Understanding a fundamental of Blockchain Technology.
2. Discuss the concept of Blockchain Technology.
3. Compare the types of Blockchain Technology.
4. Evaluate the application of Blockchain Technology

Course Contents:

Unit 1 Introduction to Blockchain [9] Disturbed Database, Advantages over conventional distributed database, What is Blockchain, Working of Blockchain, Advantages and Disadvantages of Blockchain, Merkle Tree, Gas Limit, Transactions in Blockchain, Soft and Hard Fork, SHA256 encryption
Unit 2 Blockchain types and Consensus Mechanisms[9] Public Blockchain, Private Blockchain, Hybrid Blockchain, Consortium Blockchain, Advantages and Use Cases of types of Blockchains, Proof-of-Work, Proof-of-Stake, Byzantine Fault Tolerance
Unit 3 Cryptocurrency:[9] Introduction to Cryptocurrency, Types of Cryptocurrencies, Cryptocurrency Wallets, MetaMask, Tokens and Stable Coins, Cryptocurrency Attacks, Creating MetaMask Wallet for Testnet Ethers
Unit 4 Ethereum and Smart Contracts:[9] Introduction to Ethereum, Ethereum Networks, Web3.js, Overview of Smart Contracts, Characteristics of Smart Contracts, Types of Smart Contracts, Life-cycle of a Smart Contract, Remix IDE to run Smart Contracts



Unit 5

Applications of Blockchain:[9]

Blockchain in Financial Services, Supply Chain Management, and Government Services, Blockchain in Healthcare.

Text Books:

1. Imran Bashir, "Mastering Blockchain: Distributed Ledger Technology, decentralization, and smart contracts explained", 2nd Edition, Packt Publishing Ltd, March 2018.
2. Bellaj Badr, Richard Horrocks, Xun (Brian) Wu, "Blockchain By Example: A developer's guide to creating decentralized applications using Bitcoin, Ethereum, and Hyperledger", Packt Publishing Limited, 2018.

Reference Books:

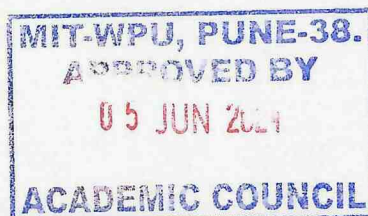
- [1] N. Daskalakis and P. Georgitseas, "An introduction to cryptocurrencies: The crypto market ecosystem," Routledge, Abingdon, Oxon, 2020.
- [2] M. G. Vigliotti and H. Jones, "The executive guide to blockchain: Using smart contracts and digital currencies in your business," Palgrave Macmillan, Cham, 2020.
- [3] D. Drescher, "Blockchain enabled applications: Understand the blockchain ecosystem and how to make it work for you," 2nd ed., Apress, [United States], 2021.

Web Resources:

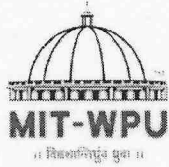
- www.blockchainbook.info
- www.blockchain-council.org
- www.investopedia.com/terms/b/blockchain.asp
- www.blockgeeks.com
- www.coindesk.com

Pedagogy:

Participative learning,
discussions, fundamental concepts,
types and application of Blockchain Technology.



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TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS

Course Structure

Course Code	BCA30120				
Course Category	Program Elective -I				
Course Title	Data Mining and Warehousing				
Teaching Scheme	Lectures	Tutorials	Laboratory/Practical	Project	Total
Weekly Load Hrs.	3	1			4
Credits	3	1			4
Assessment Schema Code	TT1				

Course Objectives:

1. To gain a comprehensive understanding of data mining and data warehousing components
 2. To develop the proficiency to apply essential data preprocessing techniques
 3. To cultivate the ability to choose suitable data mining techniques for real-world analysis problems
- To attain practical proficiency in utilizing weka tool to apply data mining techniques

Course Outcomes:

On completion of the course, student will be able to –

1. Understand the functionality of the various data mining and data warehousing components.
2. Choose and apply appropriate data preprocessing techniques.
3. Choose the appropriate data mining techniques for real day analysis problems.
4. Use weka tool to preprocess data and apply data mining techniques to large data sets to extract actionable knowledge.

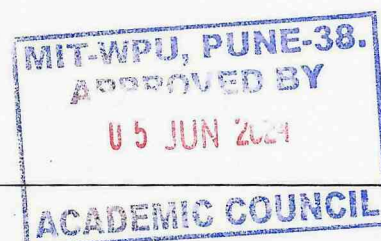
Course Contents:

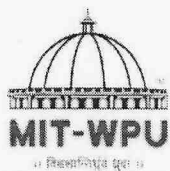
Unit 1: Introduction to Data Mining [6]

What is data mining
Types of Data to be mined
Types of Patterns can be mined
Data Mining Techniques
Knowledge Discovery in Databases
Data Mining Issues
Applications of Data Mining

Unit 2: Introduction to Data Warehouse [7]

Compare operational database and data warehouse
Need of data warehouse
Data Warehousing: A Multitiered Architecture
Data Warehouse Models: Enterprise Warehouse, Data Mart, and Virtual Warehouse
Extraction, Transformation, and Loading
Data Cube: A Multidimensional Data Model
Data Warehouse Modeling: Data Cube and OLAP
Data Warehouse Design and Usage





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Unit 3: Data pre-processing [7]

Data pre-processing: An overview

Data Cleaning

Data Integration

Data Reduction

Data Transformation and Data Discretization

Unit 4: Data Mining Techniques: Association Rule Mining [7]

Basic idea: item sets, Frequent Item-sets,

Closed Itemsets, Association Rules

Frequent Itemset Mining Methods:

Apriori Algorithm: Finding Frequent Itemsets by Confined

Candidate Generation

Generating Association Rules from Frequent Itemsets

Improving the Efficiency of Apriori

Unit 5: Data Mining Techniques: Classification [7]

Definition of Classification

General approach of classification

Decision tree Induction:

Attribute Selection Measures

Tree Pruning

Bayesian Classification: Bayes Theorem, Naïve Bayes classifier, Bayesian Networks

Unit 6: Data Mining Techniques : Clustering [6]

Cluster analysis

Overview of clustering methods

k-means

Hierarchical clustering

Unit 7: Introduction to Weka Tool [5]

Introduction

Installation

Loading data

File formats

Preprocessing the data

Execution of algorithms for Association rule mining , classification and clustering

Learning Resources:

Textbooks:

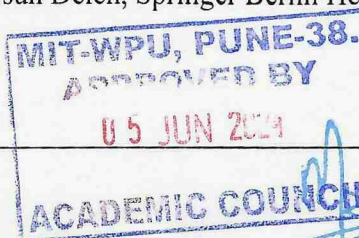
1. Data Mining: Concepts and Techniques, Jiawei Han, Micheline Kamber, Jian Pei, Elsevier Science 3rd Edition, John Wiley & Sons, 2011

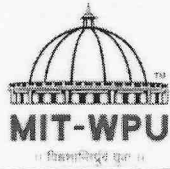
Reference Books:

1. Data Mining and Data Warehousing Principles and Practical Techniques, Parteek Bhatia, Cambridge University Press, 1st Edition, 2019
2. Advanced Data Mining Techniques, David L. Olson, Dursun Delen, Springer Berlin Heidelberg, 1st Edition, 2008

Web Links:

1. https://www.tutorialspoint.com/weka/weka_tutorial.pdf





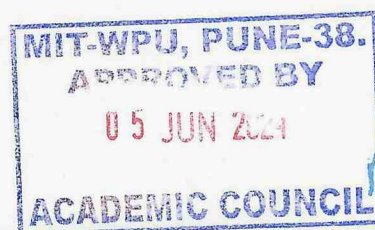
Dr. Vishwanath Karad

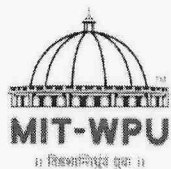
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Pedagogy:

Participative learning,
discussions,
problem solving,
assignments,
Tutorials,
Lab assignment





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TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS

Course Structure

Course Code	BCA30130				
Course Category	Program Elective -I				
Course Title	Agile Frameworks				
Teaching Scheme	Lectures	Tutorials	Laboratory/Practical	Project	Total
Weekly Load Hrs.	3	1			4
Credits	3	1			4
Assessment Schema Code	TT1				

Course Objectives:

- To understand the concept of Software Development
- To be familiar with Software engineering process

Course Outcomes:

- Students will ably differentiate between static and dynamic behavior of software development.
- Students will be able to explain various types of Agile frameworks to address the dynamic behavior of software development.
- Students will be able to differential various agile frameworks.

Course Contents:

Unit 1: Introduction to Agile software development [6]

Software Engineering process -goals and applications

Basic Concepts of Agile development,

Values and Principles of agile development

Significance of agile development

Unit 2: Kanban [8]

Principles and lifecycle of Kanban

Core practices of Kanban

Kanban Board

Roles and responsibilities in Kanban

Benefits of Kanban

Unit 3: Extreme Programming [8]

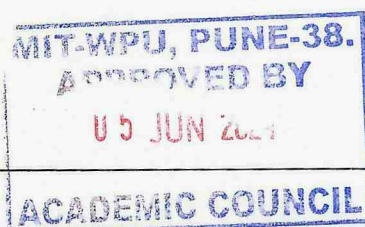
Principles of XP

lifecycle XP

Work products

Roles and responsibilities in XP

XP practices.



Unit 4: Scrum [8]

Principles of Scrum
Scrum Team
Scrum values
Scrum life cycle
three artifacts.
Scrum events

Unit 5: SaFe [8]

History of SaFe
Key principles
Best practices for SaFe implementation
Introduction about Jira tool

Unit 6: LeSS [7]

Introduction to LeSS
Rules for implementing Less in a multi team context.
Less Team structure.
Management of LeSS development life cycle.

Learning Resources:

Reference Books:

1. AGILE in Practice: Practical Use-cases on Project Management Methods including Agile, Kanban and Scrum [Edition 1]
2. Effective project management: traditional, agile, extreme by Wysocki, Robert K. John Wiley & Sons, 2011.

Supplementary Reading

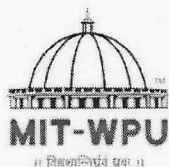
1. Learning Agile : Understanding Scrum, XP, Lean, and Kanban. Andrew Stellman and Jennifer Greene
2. Agile Project Management with Scrum, by Ken Schwaber

Web Links:

1. <https://www.udemy.com/>
2. <https://www.coursera.org/>
3. <http://www.scaledagileframework.com>
- <https://www.digite.com/kanban/what-is-kanban/>

Pedagogy:

Participative learning, discussions, PowerPoint presentation.



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COURSE STRUCTURE

Course Code	BCA30140				
Course Category	Program Elective -I				
Course Title	Introduction to NoSQL Database				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3	1	0		4
Credits	3	1	0		4
Assessment Code	TT1				

Prerequisites:

Knowledge of database concepts

Basic understanding of Big Data platforms

Course Objectives:

1. Understand importance of NoSQL Databases.

Course Outcomes:

On completion of the course, student will be able to–

- 1 Explain and compare different types of NoSQL Databases
- 2: Compare and contrast RDBMS with different NoSQL databases.
- 3: Demonstrate the detailed architecture and performance tune of Document-oriented NoSQL databases.
- 4: Explain performance tune of Key-Value Pair NoSQL databases.
- 5: Apply Nosql development tools on different types of NoSQL Databases.

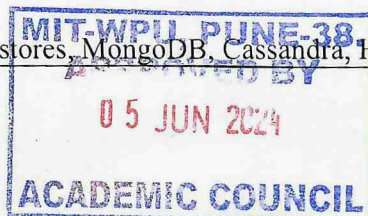
Course Contents:

Unit 1: Introduction to NoSQL [9]

Overview and History of NoSQL Databases. Definition of the Four Types of NoSQL Database, The Value of Relational Databases, Getting at Persistent Data, Concurrency, Integration, Impedance Mismatch, Application and Integration Databases, Attack of the Clusters, The Emergence of NoSQL, Key Points.

Unit 2: Types of NoSQL Databases [9]

Comparison of relational databases to new NoSQL stores, MongoDB, Cassandra, HBASE, Neo4j use and



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deployment, Application, RDBMS approach, Challenges NoSQL approach, Key-Value and Document Data Models, Column-Family Stores, Aggregate-Oriented Databases. Replication and sharding, MapReduce on databases. Distribution Models, Single Server, Sharding, Master-Slave Replication, Peer-to-Peer Replication, Combining Sharding and Replication.

Unit 3: Features and Application of NoSQL Databases [9]

NoSQL Key/Value databases using MongoDB, Document Databases, Document oriented Database Features, Consistency, Transactions, Availability, Query Features, Scaling, Suitable Use Cases, Event Logging, Content Management Systems, Blogging Platforms, Web Analytics or Real-Time Analytics, E-Commerce Applications, Complex Transactions Spanning Different Operations, Queries against Varying Aggregate Structure.

Unit 4: Column-oriented NoSQL DB [9]

Column- oriented NoSQL databases using Apache HBASE, Column-oriented NoSQL databases using Apache Cassandra, Architecture of HBASE, Column-Family Data Store Features, Consistency, Transactions, Availability, Query Features, Scaling, Suitable Use Cases, Event Logging, Content Management Systems, Blogging Platforms, Counters, Expiring Usage.

Unit 5: NoSQL Development Tools [9]

NoSQL Key/Value databases using Riak, Key-Value Databases, Key-Value Store, Key-Value Store Features, Consistency, Transactions, Query Features, Structure of Data, Scaling, Suitable Use Cases, Storing Session Information, User Profiles, Preferences, Shopping Cart Data, Relationships among Data, Multi operation Transactions, Query by Data, Operations by Sets. Graph NoSQL databases using Neo4, NoSQL database development tools and programming languages, Graph Databases, Graph Database. Features, Consistency, Transactions, Availability, Query Features, Scaling, Suitable Use Cases.

Learning Resources:

Reference Books:

Sadalage, P. & Fowler, NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence, Wiley Publications, 1st Edition, 2019.

Web Resources

Weblinks

1. <https://www.ibm.com/cloud/learn/nosql-databases>
2. <https://www.coursera.org/lecture/nosql-databases/introduction-to-nosql-VdRNp>
3. <https://www.geeksforgeeks.org/introduction-to-nosql/>
4. <https://www.javatpoint.com/nosql-databa>

MOOCs: Online courses for self-learning

Courses by NPTEL and MIT Open Courseware etc

Pedagogy:

- Participative Learning,
- Discussion
- Demonstrations
- Practical
- Assignment



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TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS

COURSE STRUCTURE

Course Code	BCA30150				
Course Category	Program Elective - I				
Course Title	Enterprise Infrastructure Security				
Teaching Scheme and Credits	L	T	Laboratory	Project	Credits
Weekly load hrs.	3	1	-		4
Credits	3	1			4

Assessment Schema Code: TT1

Pre-requisites: Need basics Java Programming, Database and Web Technology concepts

Course Objectives:

The course aims to provide students with a comprehensive understanding of enterprise infrastructure security practices, policies, and technologies. By the end of the course, students will be able to effectively design, implement, and manage security measures to safeguard enterprise infrastructure.

Learning Outcomes:

Evaluate the security risks and threats to enterprise infrastructure.

Design and implement security measures to protect enterprise networks, systems, and data.

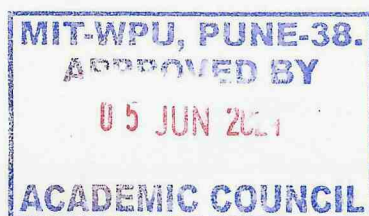
Analyze and assess the effectiveness of security controls in enterprise infrastructure.

Demonstrate the ability to respond to and mitigate security incidents in enterprise environments.

Develop policies and procedures to ensure compliance with security regulations and standards.

Course Outcomes:

1. Understand fundamentals of Big Data analytics
2. Explore the Hadoop framework and Hadoop Distributed File system
3. Illustrate the concepts of NoSQL using MongoDB and Cassandra for Big Data
4. Employ MapReduce programming model to process the big data
5. Understand various machine learning algorithms for Big Data Analytics, Web Mining and Social Network Analysis.



Course Contents:

Unit 1: : Introduction to Enterprise Infrastructure Security [5]

Overview of enterprise infrastructure security Importance of securing enterprise infrastructure Key security concepts and principles

Unit 2: Risk Management in Enterprise Infrastructure Security [5]

Understanding security risks in enterprise infrastructure

Identifying and assessing security risks

Implementing risk management strategies and mitigation techniques [5]

Unit 3: Network Security in Enterprise Infrastructure[5]

Overview of network security

Securing network devices and infrastructure

Implementing network security protocols and technologies

Unit 4: : Identity and Access Management in Enterprise Infrastructure Security [5]

Managing user identities and access control in enterprise infrastructure

Implementing authentication and authorization mechanisms

Enforcing access control policies and permissions

Unit 5: Data Security in Enterprise Infrastructure [5]

Securing sensitive data in enterprise infrastructure

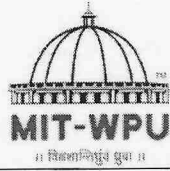
Implementing data encryption and protection mechanisms

Ensuring data integrity and confidentiality

Chapter 6: Application Security in Enterprise Infrastructure [5]

Securing enterprise applications and software

Implementing secure coding practices



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Conducting application security testing and assessments

Chapter 7: Cloud Security in Enterprise Infrastructure [5]

Understanding cloud security risks and challenges

Securing cloud-based infrastructure and services

Implementing security controls and measures in the cloud environment

Chapter 8: Incident Response and Management in Enterprise Infrastructure Security [5]

Establishing incident response procedures and protocols

Responding to security incidents and breaches in enterprise infrastructure

Implementing incident management tools and processes

Chapter 9: Compliance and Governance in Enterprise Infrastructure Security [5]

Understanding regulatory compliance requirements for enterprise infrastructure

Implementing governance frameworks and best practices

Conducting security audits and assessments

Learning Resources:

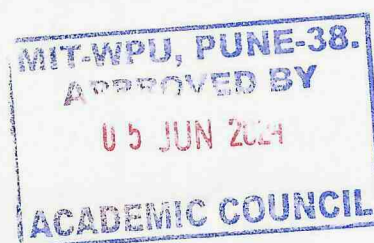
Textbook: "Enterprise Security Architecture: A Business-Driven Approach" by John Sherwood, Andrew Clark, and David Lynas

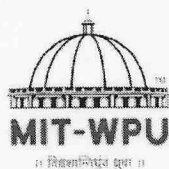
Case Studies: Real-world examples of security breaches and their impact on organizations

Online Resources: Articles, whitepapers, and research papers on enterprise infrastructure security practices

Simulation Software: Tools for hands-on experience in designing and implementing security measures

Guest Lectures: Industry experts providing insights into the latest trends and best practices in enterprise infrastructure security.





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COURSE STRUCTURE

Course Code	BCA300160			
Course Category	Program Elective-II			
Course Title	Introduction to data science			
Teaching Scheme and Credits	L	T	Lab	Credits
Weekly load hrs.	3	1		4
credits	3	1		4
Assessment Schema Code	TT1			

Pre-requisites: Basics of statistics and mathematics

Course Objectives:

Students will learn:

1. data analysis techniques and quantitative modeling for the solution of real-world business problems
2. The core competencies of data science, starting from the philosophy and ethics of data collection and analysis to data handling, to statistical modeling and scientific data visualization and communication.
3. The statistical data analysis techniques utilized in business decision making.

Course Outcomes:

1. Demonstrate and understanding of the principles of data science
2. Critically select and apply appropriate methods and techniques to extract relevant and important information from data.
3. Use appropriate modeling techniques for simple data science problems and Communicate results effectively

Course Contents

Unit 1: Introduction: 07

Introduction to Data Science Evolution of Data Science Data Science Roles– Stages in a Data Science Project Applications of Data Science in various fields Data Security Issues.

Unit 2: Data Collection and Data Pre-Processing 07

Data Collection Strategies Data Pre-Processing Overview Data Cleaning Data Integration and Transformation Data Reduction Data Discretization.

Unit 3: Exploratory Data Analytics 08

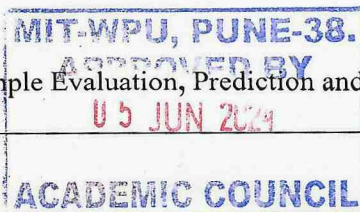
Descriptive Statistics Mean, Standard Deviation, Skewness and Kurtosis Box Plots PivotTable Heat Map Correlation Statistics ANOVA.

Unit 4: Model Development - I 08

Simple and Multiple Regression Model Evaluation using Visualization Residual Plot Distribution Plot

Unit 5: Model Development- II 08

Polynomial Regression and Pipelines, Measures for In-sample Evaluation, Prediction and Decision Making.



Unit 6: Model Evaluation 07

Generalization Error, Out-of-Sample Evaluation Metrics, Cross Validation, Over fitting, Under Fitting and Model Selection, Prediction by using Ridge Regression, Testing Multiple Parameters by using GridSearch.

Learning Resources:

Textbooks:

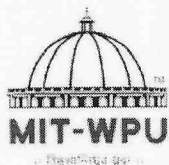
1. Data science, John D. Kelleher, Brendan Tierney, The MIT Press, Cambridge, Massachusetts, 6th Edition, 2018
2. Data Mining: Concepts and Techniques, Jiawei Han, Micheline Kamber, Jian Pei, Elsevier Science 3rd Edition, John Wiley & Sons, 2011

Reference Books:

1. R for data science, Dan Toomey, Packt Publishing, Birmingham, UK, 3rd Edition, 2014
2. Practical data science for information professionals, David Stuart, Facet, London, 7th Edition, 2020

Pedagogy:

- Participative learning
- Discussions
- problem solving assignment
- conceptual and contextual learning
- PPT Presentation & Interactive discussion



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TECHNOLOGY RESEARCH SOCIAL INNOVATION & PARTNERSHIPS

Course Structure

Course Code	BCA30170				
Course Category	Program Elective -II				
Course Title	DevOps				
Teaching Scheme	Lectures	Tutorials	Laboratory/Practical	Project	Total
Weekly Load Hrs.	3	1			4
Credits	3	1			4
Assessment Schema Code	TT1				

Course Objectives:

To understand

1. The history of Agile Software Development and DevOps
2. The practices of version control and configuration management, and practices such as Continuous Integration (CI), Continuous Delivery (CD), Continuous Testing, Continuous Deployment, Continuous Monitoring, Continuous Feedback and Continuous Improvement
3. The skills, culture, mindset and tools required to establish a DevOps ecosystem, and the deployment pipeline and its purpose
4. DevOps tools and their purpose

Course Outcomes:

After the completion of the course students will understand and apply their knowledge acquired on,

1. DevOps with reference to SDLC models including Agile
2. DevOps practices such as Configuration Management and Version Control, Continuous Integration, Continuous Deployment, and Continuous Testing
3. DevOps tools/platforms such as Git, Chef Infra, Puppet in establishing a deployment pipeline.

Course Contents:

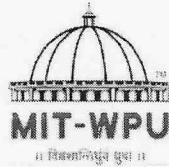
Unit 1: Linux Operating System Basics (12 sessions)

History of Unix and Linux, Difference between Unix and Linux, Difference between Linux and Windows, The structure of Linux File System, Different types of Linux shells, Installing Linux, Basic Linux commands (File operations, Directory operations, Changing file permissions and ownership, Introduction to vi or vim editor), Environment Variables, Redirection, Pipes, Filters, Types of links - soft and hard links, Introduction to Linux Services, Start and stop services, Linux processes, Find and kill a process with id and name, Packages and package management, Different types of packages (E.g. RPM and YUM)

Unit 2: Evolution of Software Engineering Methodologies (10 sessions)



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Recap of SDLC Models (Pros and Cons of different SDLC models (Waterfall Model, Incremental Process Model, Spiral Model, Component-based Model, Formal Methods), Working in a Team based Software Project (Team structure and roles, Challenges of developing software in a team, Importance of Software Engineering and SDLC models, Software Engineering tools), Roles of Business Team, Development Team, Testing Team and Operations Team.

Introduction to (or Recap of) Agile (History of Agile, Agile Manifesto, Agile Principles, Scrum, Extreme Programming, Lean)

Unit 3: The DevOps Movement (10 sessions)

History of DevOps, Definition of DevOps, Purpose of DevOps, DevOps Goals, DevOps Stakeholders, Agile and DevOps, ITIL and DevOps, Sample list of DevOps Tools, DevOps practices (Configuration management and version control, CI/CD (Continuous Integration, Continuous Delivery), Continuous Testing, Continuous Deployment, Continuous Monitoring, Continuous Feedback, Continuous Improvement), Role of DevOps Engineer, ITIL and DevOps

Unit 4: Git for Version Control & Other DevOps Tools (13 sessions)

Introduction to Version Control Systems and Types, Difference between CVCS and DVCS, Introduction to GIT, History of Git, GIT Basics - GIT Command Line, Installing Git- Installing on Linux or Windows, Initial setup, GIT Essentials- Creating repository, Cloning, Check-in and committing, Fetch, Pull and Remote branching, creating a branch, Switching and merging, Example scenarios for branching and cloning. Brief introduction to DevOps tools (such as ChefInfra, Puppet, Ansible, Terraform)

Learning Resources:

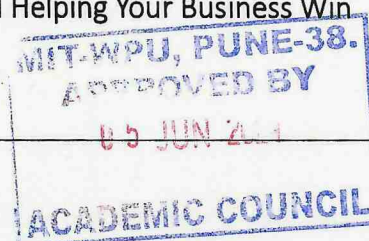
Text Books:

1. The DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations, Gene Kim, Patrick Debois, John Willis, Jez Humble, IT Revolution Press; Illustrated edition
2. Linux with Operating System Concepts, Richard Fox, Northern Kentucky University, USA

Reference Books:

1. The Phoenix Project: A Novel about IT, DevOps, and Helping Your Business Win

Web Resources:



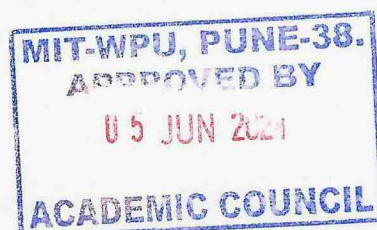
1. <https://git-scm.com/doc>
2. https://docs.chef.io/platform_overview/
3. <https://www.puppet.com/why-puppet/about-puppet>
4. <https://www.puppet.com/docs/pdf>

Weblinks:

1. <https://www.coursera.org/>
2. <https://www.atlassian.com/devops>

Pedagogy:

- Participative Learning,
- Discussion
- Demonstrations
- Practical
- Assignment



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M. Bechu

COURSE STRUCTURE

Course Code	BCA30180				
Course Category	Program Elective-II				
Course Title	Numeric and Text Analytics				
Teaching Scheme	Lecture s	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3	1			4
Credits	3	1			4
Assessment Scheme Code	TT1				

Pr-requisites:

1. Basic understanding of statistics, data analysis, and programming (preferably Python or R).

Course Objectives:

1. This course aims to provide students with the knowledge and skills necessary to analyze both numeric and text data effectively. By the end of the course, students should be able to apply analytical techniques to numeric and text datasets, extract meaningful insights, and make data-driven decisions.

Course Outcomes:

Student will able to

1. Understand the fundamentals of numeric and text analytics.
2. Analyze numeric data to identify trends, patterns, and relationships.
3. Process and analyze text data for sentiment analysis, topic modeling, and information extraction.
4. Apply statistical and machine learning techniques to numeric and text datasets.
5. Evaluate the performance of analytic models and interpret the results.
6. Demonstrate proficiency in using relevant tools and libraries for numeric and text analytics.

Course Contents:

Unit 1 : Introduction to Numeric and Text Analytics: 9

- Overview of numeric and text analytics
- Importance of data analytics in decision-making
- Data sources and challenges in numeric and text data analysis
- Ethical considerations in handling data

Unit 2 : Exploratory Data Analysis for Numeric Data: 6

- Summary statistics and visualization techniques
- Distribution analysis and outlier detection
- Correlation and regression analysis
- Hypothesis testing and confidence intervals

Unit 3 : Predictive Modeling for Numeric Data: 6

- Introduction to predictive modeling techniques
- Linear and logistic regression
- Decision trees and ensemble methods
- Model evaluation and validation techniques

Unit 4 : Feature Engineering for Numeric Data: 6

- Data preprocessing techniques (e.g., normalization, scaling)
- Feature selection and dimensionality reduction
- Handling missing data and categorical variables
- Advanced feature engineering techniques

Unit 5: Topic Modeling and Text Classification: 6

- Topic modeling techniques (e.g., Latent Dirichlet Allocation)
- Text classification using supervised learning algorithms (e.g., Naive Bayes, Support Vector Machines)
- Evaluation metrics for text classification models
- Case studies of topic modeling and text classification applications

Unit 6: Text Representation and Word Embeddings: 6

- Text representation techniques (e.g., bag-of-words, TF-IDF)
- Word embeddings (e.g., Word2Vec, GloVe)

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M. Beethu

- Using pre-trained word embeddings for text analysis tasks
- Case studies of text representation and word embeddings in practice

Unit 7: Text Analytics Applications: 6

- Text summarization and document clustering
- Text similarity and document similarity measures
- Text analytics for social media and customer feedback analysis
- Case studies and real-world examples of text analytics applications

Learning Resources:

Text Books/Reference Books:

1. "Introduction to Statistical Learning" by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani.
2. "Applied Text Analysis with Python: Enabling Language-Aware Data Products with Machine Learning" by Benjamin Bengfort, Tony Ojeda, and Rebecca Bilbro.
3. "Text Mining with R: A Tidy Approach" by Julia Silge and David Robinson.

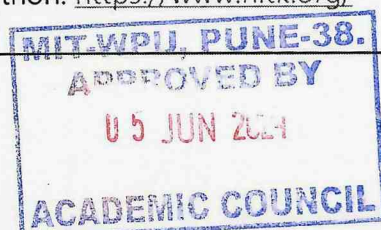
Supplementary Reading:

1. "Text Analytics with Python: A Practical Real-World Approach to Gaining Actionable Insights from your Data" by Dipanjan Sarkar.
2. "Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems" by Sowmya Vajjala, Bodhisattwa Majumder, and Anuj Gupta.
3. "Text Analysis with R for Students of Literature" by Matthew Jockers.

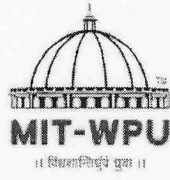
Web Resources :

Weblinks:

1. Kaggle: Offers datasets, competitions, and kernels (code notebooks) for practicing numeric and text analytics. <https://www.kaggle.com/>
2. Towards Data Science: Provides articles, tutorials, and case studies on data science and machine learning, including topics related to numeric and text analytics. <https://towardsdatascience.com/>
3. Natural Language Toolkit (NLTK) Documentation: Offers tutorials and documentation for using NLTK, a popular library for natural language processing in Python. <https://www.nltk.org/>



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Dr. Karad

Course Code	BCA30190				
Course Category	Program Elective - II				
Course Title	Application and Web application Security				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3	1	-	-	4
Credits	3	1	-	-	4
Assessment Scheme Code	TT1				

Pre-requisites:

Basic knowledge of programming languages (e.g. Java, Python, PHP)
 Basic understanding of web development technologies (e.g. HTML, CSS, JavaScript)
 Basic knowledge of networking and information security concepts

Course Objective:

1. This module provides an introduction to the principles and practices of securing applications and web applications.
2. Students will learn about common security threats and vulnerabilities in applications and web applications, and how to address them through secure coding practices and secure design principles.
3. The module will cover topics such as input validation, authentication and authorization, session management, encryption, secure communication protocols, and security testing techniques.
4. Students will also learn about common security frameworks and tools used to secure applications and web applications.

Course Outcome:

1. Identify common security threats and vulnerabilities in applications and web applications.
2. Apply secure coding practices and secure design principles to address security issues in applications and web applications.
3. Implement security controls for input validation, authentication and authorization, session management, encryption, and secure communication protocols.
4. Use security testing techniques to identify and mitigate security risks in applications and web applications.
5. Evaluate and select appropriate security frameworks and tools to secure applications and web applications.

Course Contents:

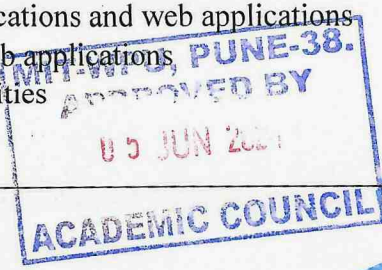
Unit 1: Introduction to Application and Web application Security:

Overview of security threats faced by applications and web applications
 Importance of securing applications and web applications
 Understanding common security vulnerabilities

Unit 2: Secure Design Principles:

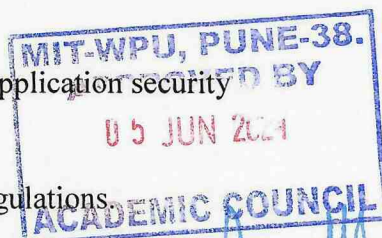
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Principles of secure application and web application design	
Secure coding practices	
Secure communication protocols	
Input validation techniques	
Unit 3: Authentication and Authorization:	5
Importance of authentication and authorization in application security	
Different authentication methods and best practices	
Role-based access control	
Single sign-on solutions	
Unit 4: Session Management:	3
Importance of secure session management	
Session fixation and session hijacking attacks	
Best practices for session management	
Cookie security	
Unit 5: Cross-Site Scripting (XSS) and Cross-Site Request Forgery (CSRF):	3
Understanding XSS and CSRF attacks	
Preventing XSS and CSRF attacks	
Best practices for mitigating XSS and CSRF vulnerabilities	
Unit 6: SQL Injection and Injection Attacks:	5
Understanding SQL injection and other injection attacks	
Preventing SQL and other injection attacks	
Best practices for secure coding to prevent injection vulnerabilities	
Unit 7: Security Testing and Vulnerability Assessment:	5
Importance of security testing in application development	
Common security testing tools and techniques	
Conducting vulnerability assessments and penetration testing	
Reporting and remediation of security vulnerabilities	
Unit 8: Secure Development Lifecycle:	5
Introduction to secure development lifecycle models	
Incorporating security into the software development process	
Secure coding standards and best practices	
Security training and awareness for developers	
Unit 9: Security in Mobile Applications:	5
Security considerations for mobile applications	
Secure coding practices for mobile applications	
Mobile application security testing	
Best practices for securing mobile applications	
Unit 10: Secure Deployment and Maintenance:	3
Secure deployment strategies for applications and web applications	
Patch management and updates	
Monitoring and incident response	
Secure configuration management practices	
Unit 11: Legal and Compliance Considerations:	2
Understanding legal and compliance requirements for application security	
Data protection regulations (e.g., GDPR, HIPAA)	
Industry-specific compliance standards (e.g., PCI DSS)	
Best practices for ensuring compliance with security regulations	



Learning Resources:

Textbooks/Reference Books:

1. "The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws" by Dafydd Stuttard and Marcus Pinto
2. "OWASP Testing Guide v4" by OWASP
3. "Web Application Security: A Beginner's Guide" by Bryan Sullivan and Vincent Liu

Web Resources

Weblinks:

1. <https://www.coursera.org/>
2. <https://www.udemy.com/>
3. <https://www.pluralsight.com/>

Pedagogy:

- Participative Learning,
- Discussion
- Demonstrations
- Practical
- Assignment

MOOCs: Online courses for self-learning

Courses by NPTEL and MIT Open Courseware etc

Pedagogy:

- Participative Learning,
- discussions,
- problem solving,
- assignments,



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- A signature that appears to be "B. K. Kulkarni" in blue ink.
- Another signature in blue ink, possibly "M. Becham".

Course Structure

Course Code	BCA30200				
Course Category	Program Elective-III				
Course Title	Data Visualization using Tableau				
Teaching Scheme	Lectures	Tutorials	Laboratory/Practical	Project	Total
Weekly Load Hrs.	3	1			4
Credits	3	1			4
Assessment Schema Code	TT1				

Pre-requisites:

Basic knowledge of data analytics, statistics and Mathematics

Course Objectives:

The main objectives of the course are as follows:

1. Understand and describe the main concepts of data visualization
2. Understand the main chart types and their recommended usage
3. Create ad-hoc reports, data visualizations, and dashboards using Tableau Desktop

Course Outcomes:

At the end of the course, students will be able to

1. Able to understand and describe the main concepts of data visualization
2. Able to understand the main chart types and their recommended usage
3. Able to create ad-hoc reports, data visualizations, and dashboards using Tableau Desktop

Course Contents:

Unit 1: Introduction to Data Visualization [9]

- Course introduction
- Intro to data visualization
- Getting started with Tableau Desktop
- Creating the first charts

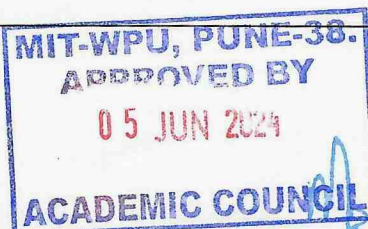
Common charts

- Creating common visualizations (bar charts, line charts etc.)
- Filtering and sorting data

Adding Titles, Labels, and description

Unit 2: Interactions [9]

- Dataviz best practices (lecture)
- Interactivity with text and visual tooltips
- Interactivity with actions (filter, highlight, URL)



Assembling dashboards from multiple charts

Unit 3: Transform the data [9]

- Creating simple calculations in Tableau
- Using table calculations
- Intro to Tableau Prep

Manipulating Data in Tableau

- Cleaning-up the data with the Data
- Structuring your data
- Sorting and filtering Tableau data

Pivoting Tableau data

Unit 4: Advanced visualizations [9]

- Using Filters
- Using the Detail panel
- Using the Size panels
- Customizing filters
- Using and Customizing tooltips
- Formatting your data with colors
- Working with more advanced chart types

Working with geospatial data and maps

Unit 5: Distributing & Publishing Your Visualization [9]

- Tableau file types
- Publishing to Tableau Online
- Sharing your visualization

Printing and exporting

Learning Resources:

Text Books/Reference Books::

- Tableau 9: The Official Guide, George Peck: 9780071816786
- "Tableau 10 Complete Reference", by Joshua N Milligan
- The Big Book of Dashboards: Visualizing Your Data Using Real-World Business Scenarios · Steve Wexler
- Tableau Desktop Pocket Reference: Essential Features, Syntax, and Data Visualizations : Sleeper, Ryan

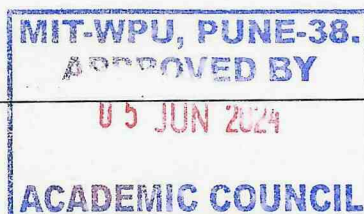
Web Resources

Weblinks:

https://www.tableau.com/learn/training/elearning?&d=7013y000000vYhH&gad_source=1&gclid=CjwKCAjwupGyBhBBEiwA0UcqaOVS5rn8nrGFtkjffYwUe_9pFvofv5uedvpr9SU-Vpxcsya5lGGLmRoCokEQAvD_BwE&gclsrc=aw.ds

Pedagogy:

- Participative Learning,



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- Discussion
- Demonstrations
- Practical
- Assignment

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SP. Bhole.

COURSE STRUCTURE

Course Code	BCA30210				
Course Category	Program Elective-III				
Course Title	Testing and Automation-I				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3	1	0		4
Credits	3	1	0		4
Assessment Code	TT1				

Prerequisites:

Knowledge about software and software engineering

Course Objectives:

Students will learn:

1. Basic software debugging methods.
2. White box testing methods and techniques.
3. Black Box testing methods and techniques.
4. Designing test plans.
- 5 Different testing tools (familiar with open source tools)

Course Outcomes:

On completion of the course, student will be able to–

1. Investigate the reason for bugs and analyze the principles in software testing to prevent and remove bugs.
2. Implement various test processes for quality improvement
3. Design test planning
4. Manage the test process

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5. Apply the software testing techniques in commercial environment

6. Use practical knowledge of a variety of ways to test software and an understanding of some of the tradeoffs between testing techniques

Course Contents:

Unit 1: Software Quality Assurance [9]

Software Quality Assurance, Software Quality, Quality Control, Quality Assurance, Quality Factors, Difference between Quality control and Quality assurance, Quality Movement,

Background issues and SQA activities Software Reviews

Formal Technical Reviews, Software Reliability, SQA Plan, The ISO 9001 Quality Standard

Unit 2: Software Quality Improvement [9]

Introduction, Pareto Diagrams, Cause-effect Diagrams, Scatter Diagrams, Run charts

Unit 3: Introduction to Software testing [9]

Software Testing, Terms: fault, failure, error, fault masking, test, test case, Fundamental Test process: test planning, test specification, test execution, test records, test completion, Difference between QA and Testing, Validation & Verification

Unit 4: Software Testing Methods [9]

Strategic Approach to Software Testing, Unit Testing, Integration Testing, Validation Testing, System Testing, Acceptance testing

Unit 5: Software Testing Tools [9]

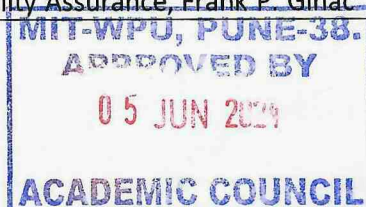
Develop test cases for manual testing. Introduction to automation testing tool: JMeter

Jmeter case study.

Learning Resources:

Reference Books:

1. Testing Computer Software, CemKaner, Jack Falk, and Hung Quoc Nguyen
2. Practical Software Testing: A Process-Oriented Approach, Burnstein, Springer, ISBN 978-81-8128-089-3
3. Software Testing in the Real World: Improving the Process, Edward Kit
4. The Art of Software Testing, Glenford J. Myers, Wiley.
5. Customer Oriented Software Quality Assurance, Frank P. Ginac



m. Beckman
Reale

6. Metrics and Models in Software Quality Engineering, Stephen H. Kan,
 7. Software testing by yogesh singh Cambridge publication
- Quality Management, 5th ed., Prentice-Hall, 2010. Donna C. S. Summers

Web Resources

Weblinks:

<https://www.geeksforgeeks.org/software-testing-tutorial/>

<https://www.javatpoint.com/software-testing-tutorial>

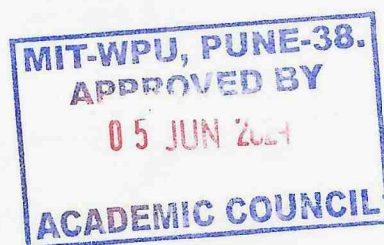
https://www.tutorialspoint.com/software_testing/index.htm

MOOCs: Online courses for self-learning

Courses by NPTEL and MIT Open Courseware etc

Pedagogy:

- Participative Learning,
- Discussion
- Demonstrations
- Practical
- Assignment



Beck

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COURSE STRUCTURE

Course Code	BCA30220				
Course Category	Program Elective-III				
Course Title	Healthcare and Finance Analytics				
Teaching Scheme	Lectur es	Tutorial s	Laboratory / Practical	Proje ct	Tot al
Weekly load hours	3	1			4
Credits	3	1			4
Assessment Scheme Code	TT1				

Pr-requisites:

1. Basic understanding of statistics, data analysis, and familiarity with healthcare and financial concepts.

Course Objectives:

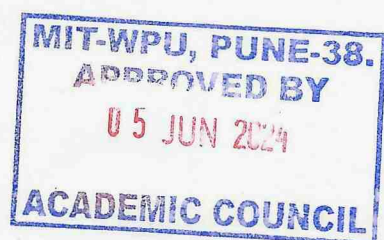
2. This course aims to equip students with the knowledge and skills necessary to analyze healthcare and financial data effectively. By the end of the course, students should be able to apply analytical techniques to healthcare and financial datasets, derive actionable insights, and make informed decisions in healthcare and financial contexts.

Course Outcomes:

Student will able to

7. Understand the fundamentals of healthcare and financial data analytics.
8. Analyze healthcare data to identify trends, patterns, and anomalies.
9. Apply statistical and machine learning techniques to predict healthcare outcomes and financial trends.
10. Evaluate the financial performance of healthcare organizations and make data-driven recommendations.

11. Interpret and communicate analytical findings to stakeholders in healthcare and financial sectors.



M. Beekun
EP. Beekun.

Course Contents:

Unit 1 : Introduction to Healthcare and Financial Analytics: 9

- Overview of healthcare and financial analytics
- Importance of data analytics in healthcare and finance
- Data sources and challenges in healthcare and financial data analysis
- Ethical considerations in handling healthcare and financial data

Unit 2 : Healthcare Data Analysis: 6

- Types of healthcare data (e.g., electronic health records, claims data)
- Data preprocessing and cleaning techniques
- Descriptive analytics in healthcare (e.g., patient demographics, disease prevalence)
- Healthcare data visualization techniques

Unit 3 : Predictive Analytics in Healthcare: 6

- Introduction to predictive modeling techniques
- Predictive modeling for healthcare outcomes (e.g., disease diagnosis, readmission prediction)
- Risk stratification and patient segmentation
- Evaluation metrics for predictive models in healthcare

Unit 4 : Healthcare Fraud Detection: 6

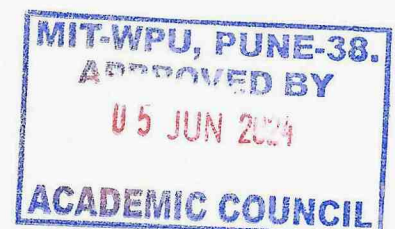
- Overview of healthcare fraud and abuse
- Techniques for detecting fraudulent activities in healthcare claims
- Anomaly detection and outlier analysis
- Case studies of healthcare fraud detection using analytics

Unit 5: Financial Data Analysis: 6

- Types of financial data (e.g., balance sheets, income statements, market data)
- Financial data preprocessing and normalization
- Financial ratio analysis for performance evaluation
- Time series analysis and forecasting in finance

Unit 6: Financial Risk Management: 6

- Introduction to financial risk management concepts
- Value at Risk (VaR) and risk modeling techniques
- Credit risk analysis and default prediction
- Stress testing and scenario analysis for financial risk assessment



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Unit 7: Healthcare Financial Analytics: 6

- Financial performance metrics for healthcare organizations (e.g., revenue cycle management, profitability analysis)
- Cost-effectiveness analysis in healthcare
- Healthcare investment analysis and ROI evaluation
- Case studies of financial analytics applications in healthcare settings

Learning Resources:

Text Books/Reference Books:

1. "Healthcare Analytics for Quality and Performance Improvement" by Trevor L. Strome.
2. "Healthcare Data Analytics" by Chandan K. Reddy, Charu C. Aggarwal, and Hui Xiong.
3. "Financial Analytics with R: Building a Laptop Laboratory for Data Science" by Mark J. Bennett and Dirk L. Hugen.

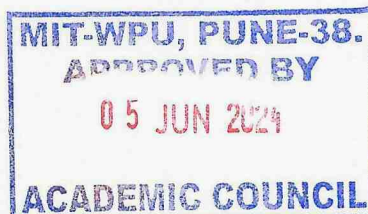
Supplementary Reading:

1. "Healthcare Analytics Made Simple: Techniques in Healthcare Computing Using Python" by Pradeep Menon and Abhinav Sagar.
2. "Financial Modeling and Valuation: A Practical Guide to Investment Banking and Private Equity" by Paul Pignataro.
3. "Healthcare Analytics: From Data to Knowledge to Healthcare Improvement" by Laura A. Kirkwood and Julie A. Garrison.

Web Resources :

Weblinks:

1. Healthcare Financial Management Association (HFMA): Provides resources, webinars, and publications related to healthcare finance and analytics. <https://www.hfma.org/>
2. HealthIT.gov: Offers information and resources on healthcare information technology, including data analytics and interoperability. <https://www.healthit.gov/>
3. Healthcare Financial Management Association (HFMA) Healthcare Analytics Online: Provides online courses and webinars on healthcare analytics topics. <https://www.hfma.org/education/online-courses/hfma-healthcare-analytics-online.html>



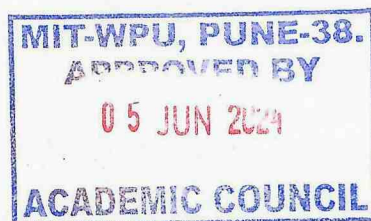
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MOOCs: Online courses for self-learning

Courses by NPTEL and MIT Open Courseware etc

Pedagogy:

- Participative Learning,
- discussions,
- problem solving,
- assignments,



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EP. Beek

COURSE STRUCTURE

Course Code	BCA30230				
Course Category	Program Elective-III				
Course Title	Cloud Security				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3	1			4
Credits	3	1			4
Assessment Scheme Code	TT1				

Pr-requisites:

1. Basic understanding of networking and cybersecurity concepts.

Course Objectives:

3. This course aims to provide students with a comprehensive understanding of cloud security principles, practices, and technologies. By the end of the course, students should be able to identify common cloud security threats, apply appropriate security measures in cloud environments, and design secure cloud architectures.

Course Outcomes:

Student will able to

1. Understand the fundamentals of cloud computing and its security implications.
2. Identify common threats and vulnerabilities in cloud environments.
3. Implement security controls and best practices to mitigate risks in cloud deployments.
4. Evaluate different cloud security solutions and select appropriate tools for specific use cases.
5. Design and deploy secure cloud architectures that meet organizational requirements.
6. Demonstrate proficiency in securing cloud-based applications and data.

Course Contents:

Unit 1 : Introduction to Cloud Computing Security: 6

- Overview of cloud computing models (IaaS, PaaS, SaaS)
- Cloud deployment models (public, private, hybrid)
- Security challenges and considerations in cloud environments
- Shared responsibility model in cloud security

Unit 2 : Cloud Security Threats and Vulnerabilities: 6

- Common threats to cloud infrastructure and services
- Data breaches, data loss, and data leakage in the cloud
- Insider threats and privilege escalation
- Attacks on cloud APIs and interfaces
- Case studies of notable cloud security incidents

Unit 3 : Cloud Security Controls and Best Practices: 6

- Identity and access management (IAM) in the cloud
- Encryption and key management for data protection
- Network security controls (firewalls, VPNs, etc.) in cloud environments
- Security monitoring and incident response in the cloud

Unit 4 : Secure Cloud Architectures: 7

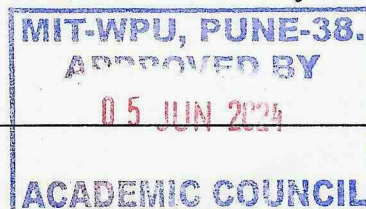
- Design principles for secure cloud architectures
- Security considerations for cloud migration and hybrid cloud deployments
- Multi-tenancy and isolation techniques
- Case studies of secure cloud architectures

Unit 5: Compliance and Legal Aspects of Cloud Security: 6

- Regulatory compliance requirements (e.g., GDPR, HIPAA) for cloud deployments
- Legal considerations and contractual agreements (SLAs) in cloud computing
- Risk management frameworks and assessment methodologies for cloud security

Unit 6: Cloud Security Tools and Technologies: 7

- Overview of cloud security platforms and solutions
- Security automation and orchestration in the cloud
- Cloud-native security services (AWS Security Hub, Azure Security Center, etc.)
- Third-party security tools and integrations



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Bottom right: *M. B. B.*

Unit 7: Secure Development and Deployment in the Cloud: 7

- Secure coding practices for cloud-based applications
- DevSecOps principles and practices
- Continuous security testing and vulnerability management
- Secure deployment strategies (container security, serverless security, etc.)

Learning Resources:

Text Books/Reference Books:

1. "Cloud Security and Privacy: An Enterprise Perspective on Risks and Compliance" by Tim Mather, Subra Kumaraswamy, and Shahed Latif.
2. "Cloud Computing Security: Foundations and Challenges" by John R. Vacca.
3. "Architecting Secure Cloud Systems: A Guide for Systems Architects" by Kevin L. Jackson and Scott Goessling.

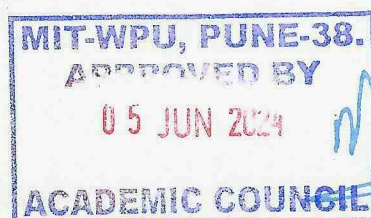
Supplementary Reading:

1. Whitepapers and research papers on cloud security from reputable sources such as Gartner, Forrester, and IDC.
2. Security blogs and forums focused on cloud security topics, such as CloudSecList and Reddit's r/cloudsecurity.
3. Hands-on labs and tutorials available on platforms like AWS Labs, Azure Labs, and Google Cloud Labs.

Web Resources :

Weblinks:

1. Cloud Security Alliance (CSA): <https://cloudsecurityalliance.org/>
2. OWASP Cloud Security Project: https://www.owasp.org/index.php/OWASP_Cloud_Security_Project
3. NIST Special Publication 800-144: <https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-144.pdf>

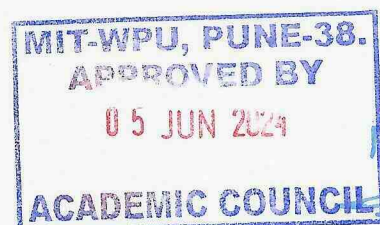


MOOCs: Online courses for self-learning

Courses by NPTEL and MIT Open Courseware etc

Pedagogy:

- Participative Learning,
- discussions,
- problem solving,
- assignments,



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COURSE STRUCTURE

Course Code	BCA30240				
Course Category	Program Elective-IV				
Course Title	Deep Learning using Python				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3	1	-	-	4
Credits	3	1	-	-	4
Assessment Scheme Code	TT1				

Pre-requisites:

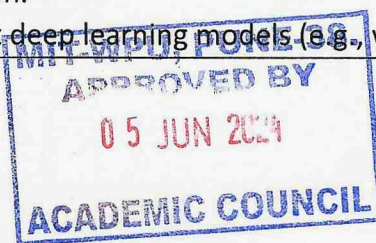
Programming in PYTHON

Description:

Deep learning has resurged with the availability of massive datasets and affordable computing, enabling new applications in computer vision and natural language processing. This course introduces convolutional, recurrent, and other neural network architectures for deep learning. Students design, implement, and train these models to solve real-world problems.

Learning Outcome:

1. Solve problems in probability and machine learning.
2. Evaluate, in the context of a case study, the advantages and disadvantages of deep learning neural network architectures and other approaches.
3. Implement deep learning models in Python using the TensorFlow and Keras libraries and train them with real-world datasets.
4. Design convolution networks for handwriting and object classification from images.
5. Design recurrent neural networks with attention mechanisms for natural language classification, generation, and translation.
6. Evaluate the performance of different deep learning models (e.g., with respect to the bias-



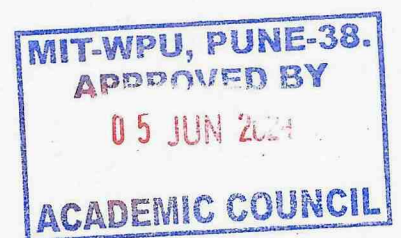
variance trade-off, overfitting and underfitting, estimation of test error).

7. Perform regularization, training optimization, and hyperparameter selection on deep models.

8. Analyze a deep learning model in preparation for deployment.

Course outcome:

1. Data Literacy - Recognize that data can have value and play a key role in society by providing opportunities to expand knowledge, to innovate, and to influence.
2. Data Science - Organize, visualize, and analyze large, complex datasets using descriptive statistics and graphs to make decisions.
3. Data Science - Identify, assess, and select appropriately among data analytics methods and models for solving real-world problems, weighing their advantages and disadvantages.
4. Data Science - Understand data science concepts, techniques, and tools to support big data analytics.



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Unit 1: Introduction to Deep Learning: 5

What is:

Deep Learning?

Artificial Intelligence (AI)?

Machine Learning (ML)?

Learning Rules & Data Representation

Unit 2: Python Programming Review: 5

Intensive review of Python programming

Introduction to Lambda Functions

Using Google Collaboratory

Unit 3: The Mathematical build of Neural Networks: 5

A First Neural Network

Data Representations

Unit 4: Introduction to TensorFlow and Keras (Part 1): 5

The definition of TensorFlow and Keras

First Step with TensorFlow

Unit 5: Introduction to TensorFlow and Keras (Parts 2): 5

First Step with Keras

Layers

Choosing a Loss Function

The fit() Method

Validation Data

Inference

Unit 6: Neural Networks: Classification: 7

Binary Classification

Multiclass Classification

Unit 7: Neural Networks: Regression: 7

Data Preparation

Building a first Model

K-fold validation

Generating predictions on new data

Unit 8: Fundamentals of machine learning: 6

The goal of Machine Learning

Evaluating Machine Learning Models

Model Fit and Generalization

MIT-WPU, PUNE-38.
APPROVED BY
05 JUN 2021
ACADEMIC COUNCIL

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J. B. B. B.

COURSE STRUCTURE

Course Code	BCA30250				
Course Category	Program Elective -IV				
Course Title	Testing and Automation-II				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3	1	0		4
Credits	3	1	0		4
Assessment Code	TT1				

Prerequisites:

Knowledge about software and software engineering

Course Objectives:

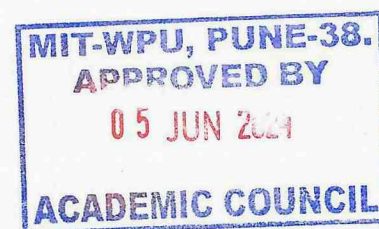
Students will learn:

1. Basic software debugging methods.
2. White box testing methods and techniques.
3. Black Box testing methods and techniques.
4. Designing test plans.
- 5 Different testing tools (familiar with open source tools)

Course Outcomes:

On completion of the course, student will be able to—

1. Investigate the reason for bugs and analyze the principles in software testing to prevent and remove bugs.
2. Implement various test processes for quality improvement
3. Design test planning
4. Manage the test process
5. Apply the software testing techniques in commercial environment
6. Use practical knowledge of a variety of ways to test software and an understanding of some of the tradeoffs between testing techniques



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Course Contents:

Unit 1: Unit Software Testing approaches 12

Black Box Testing

- Introduction,
- Need of black box testing,
- Black box testing Concept,
- Boundary Value Analysis ,
- Decision Table Testing ,
- State Transition Testing ,
- Use Case Testing ,
- Error Guessing,
- Documentation Testing and Domain Testing,
- design of test cases
- Case studies of Black-Box testing.
- Equivalence Partitioning

White Box Testing

- Introduction,
- Need of white box testing,
- Test adequacy criteria, static testing by humans,
- Basis path testing,
- Graph metrics,
- Loop Testing,
- Data flow testing,
- Control Flow Testing,
- Mutation Testing,
- Design of test cases,
- Challenges in White box testing,
- Case-study of White-Box testing.
- Experience-based Test Techniques –
- Exploratory Testing ,
- Checklist-based Testing

Grey Box Testing

Unit 2: Software Testing Methods and Strategy 12

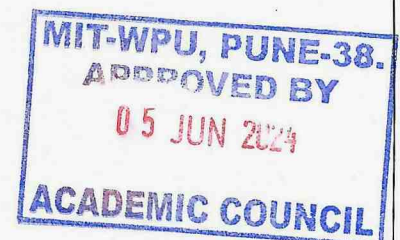
Software Testing Methods

Functional Testing

- Unit Testing,
- Integration Testing,
- System Testing
- Acceptance

Non-Functional Testing

- Performance Testing
- Security Testing
- Usability Testing



M. Bechu
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Compatibility Testing
Validation Testing,

Unit 3: Advanced Techniques and Automation Testing 12

- Manual testing Vs Automated testing,
- Automated testing tools - features, selection, mobile testing,

testing tools like QTP, Rational Robot, Winrunner, Loadrunner, selenium, Junit, monkey talk

Unit 4: Case Study- Selenium 09

Introduction to automation testing tool: Selenium

Learning Resources:

Reference Books:

8. Testing Computer Software, Cem Kaner, Jack Falk, and Hung Quoc Nguyen2.
 9. Practical Software Testing: A Process-Oriented Approach, Burnstein, Springer, ISBN 978-81-8128-089-3
 10. Software Testing in the Real World: Improving the Process, Edward Kit
 11. The Art of Software Testing, Glenford J. Myers, Wiley.
 12. Customer Oriented Software Quality Assurance, Frank P. Ginac
 13. Metrics and Models in Software Quality Engineering, Stephen H. Kan,
 14. Software testing by yogesh singh Cambridge publication
- Quality Management, 5th ed., Prentice-Hall, 2010. Donna C. S. Summers

Web Resources

Weblinks:

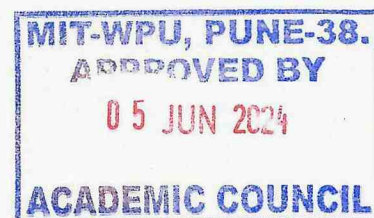
<https://www.geeksforgeeks.org/software-testing-tutorial/>
<https://www.javatpoint.com/software-testing-tutorial>
https://www.tutorialspoint.com/software_testing/index.htm

MOOCs: Online courses for self-learning

Courses by NPTEL and MIT Open Courseware etc

Pedagogy:

- Participative Learning,
- Discussion
- Demonstrations
- Practical
- Assignment



M. Beek
27/6/24

COURSE STRUCTURE

Course Code	BCA30260				
Course Category	Program Elective-IV				
Course Title	Big Data Analytics				
Teaching Scheme and Credits	L	T	Laboratory	Project	Credits
Weekly load hrs.	3	1			4
Credits	3	1			4

Assessment Scheme Code:-TT1

Pre-requisites: Need basics Java Programming, Database and Web Technology concepts

Course Objectives:

Knowledge: To understand java programming concepts with respect to

- i) Server-Side Programming
- ii) Web Development

Skills: Design skills of

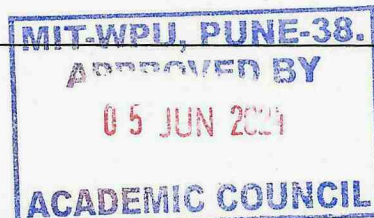
- i) Web Base Application
- ii) Distributed Application

Attitude: To develop following

- i) Programming Skills
- ii) Build Web Application in Complex situation

Course Outcomes:

6. Understand fundamentals of Big Data analytics
7. Explore the Hadoop framework and Hadoop Distributed File system
8. Illustrate the concepts of NoSQL using MongoDB and Cassandra for Big Data
9. Employ MapReduce programming model to process the big data
10. Understand various machine learning algorithms for Big Data Analytics, Web Mining and Social Network Analysis.



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Course Contents:

Unit 1: 5

Introduction to Big Data Analytics: Big Data, Scalability and Parallel Processing, Designing Data Architecture, Data Sources, Quality, Pre-Processing and Storing, Data Storage and Analysis, Big Data Analytics Applications and Case Studies.

Unit 2: 10

Introduction to Hadoop (T1): Introduction, Hadoop and its Ecosystem, Hadoop Distributed File System, MapReduce Framework and Programming Model, Hadoop Yarn, Hadoop Ecosystem Tools.

Hadoop Distributed File System Basics (T2): HDFS Design Features, Components, HDFS User Commands.

Unit 3: 10

NoSQL Big Data Management, MongoDB and Cassandra: Introduction, NoSQL Data Store, NoSQL Data Architecture Patterns, NoSQL to Manage Big Data, Shared-Nothing Architecture for Big Data Tasks, MongoDB, Databases, Cassandra Databases.

Unit 4: 10

MapReduce, Hive and Pig: Introduction, MapReduce Map Tasks, Reduce Tasks and MapReduce Execution, Composing MapReduce for Calculations and Algorithms, Hive, HiveQL, Pig.

Unit 5: 10

Machine Learning Algorithms for Big Data Analytics: Introduction, Estimating the relationships, Outliers, Variances, Probability Distributions, and Correlations,

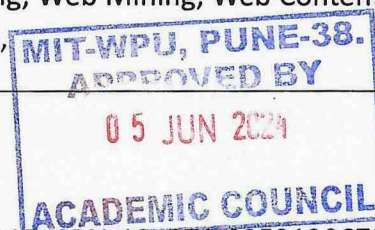
Regression analysis, Finding Similar Items, Similarity of Sets and Collaborative Filtering, Frequent Itemsets and Association Rule Mining.

Text, Web Content, Link, and Social Network Analytics: Introduction, Text mining, Web Mining, Web Content and Web Usage Analytics, Page Rank, Structure of Web and analyzing a Web Graph,

Learning Resources:

Reference Books:

1. Tom White, "Hadoop: The Definitive Guide", 4th Edition, O'Reilly Media, 2015. ISBN-13: 978-9352130672



M. Bedekar
P. Bedekar

2. Boris Lubinsky, Kevin T Smith, Alexey Yakubovich, "Professional Hadoop Solutions", 1st Edition, Wrox Press, 2014 ISBN-13: 978-8126551071

3. Eric Sammer, "Hadoop Operations: A Guide for Developers and Administrators", 1st Edition, O'Reilly Media, 2012. ISBN-13: 978-9350239261

4. Arshdeep Bahga, Vijay Madisetti, "Big Data Analytics: A Hands-On Approach", 1st Edition, VPT Publications, 2018. ISBN-13: 978-0996025577

Supplementary Reading:

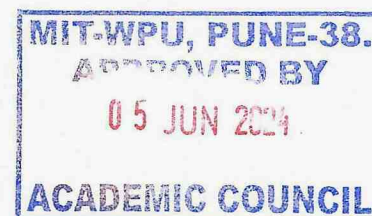
2. 1. Big Data Analytics Introduction to Hadoop, Spark, and Machine-Learning

2. "Hadoop 2 Quick-Start Guide: Learn the Essentials of Big Data Computing in the Apache Hadoop 2 Ecosystem, " Douglas Eadline,

MOOCS : Spoken Tutorial, Khan Academy

pedagogy:

- Practical learning,
- project handling,
- problem solving assignment,
- conceptual and contextual learning,
- PPT Presentation & Interactive discussion



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COURSE STRUCTURE

Course Code	BCA30270				
Course Category	Program Elective-IV				
Course Title	Ethical Hacking				
Teaching Scheme	Lectur es	Tutorial s	Laboratory / Practical	Proje ct	Tot al
Weekly load hours	3	1			4
Credits	3	1			4
Assessment Scheme Code	TT1				

Pr-requisites:

1. Basic understanding of networking concepts and cybersecurity fundamentals.

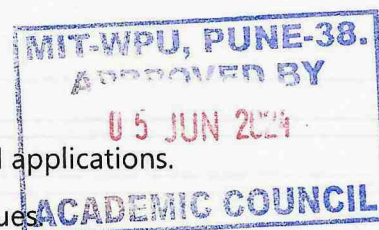
Course Objectives:

4. This course aims to provide students with the knowledge and skills necessary to identify security vulnerabilities, perform penetration testing, and strengthen the security posture of organizations ethically. By the end of the course, students should be able to conduct ethical hacking activities responsibly and effectively.

Course Outcomes:

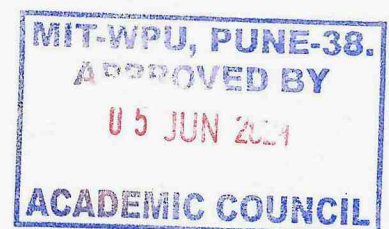
Student will able to

1. Understand the ethical and legal considerations of hacking activities.
2. Identify common security vulnerabilities in computer systems, networks, and applications.
3. Perform penetration testing using various hacking tools and techniques.
4. Analyze and interpret the results of security assessments to provide actionable recommendations.



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5. Develop strategies to secure systems and networks against cyber threats based on ethical hacking principles.



M. Bechar
sp. Bechar

Course Contents:

Unit 1 : Introduction to Ethical Hacking: 9

- Definition of ethical hacking and its importance in cybersecurity
- Ethical and legal considerations in ethical hacking
- Common hacking terminology and concepts
- Overview of hacking methodologies (e.g., reconnaissance, scanning, exploitation, post-exploitation)

Unit 2 : Information Gathering and Footprinting: 6

- Passive and active reconnaissance techniques
- Using open-source intelligence (OSINT) tools for gathering information
- Footprinting and fingerprinting of networks and systems
- Case studies of real-world reconnaissance activities

Unit 3 : Scanning and Enumeration: 6

- Network scanning techniques (e.g., port scanning, service identification)
- Host discovery and enumeration
- Vulnerability scanning and assessment
- Analyzing scan results and identifying potential security vulnerabilities

Unit 4 : Exploitation and Post-Exploitation: 6

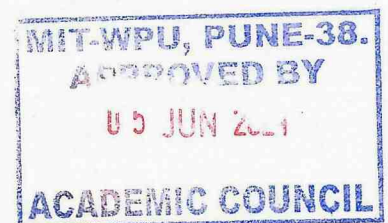
- Exploiting vulnerabilities in systems and applications
- Common attack vectors (e.g., buffer overflow, SQL injection, XSS)
- Privilege escalation techniques
- Post-exploitation activities (e.g., maintaining access, covering tracks)

Unit 5 : Wireless Network Security: 6

- Wireless network fundamentals (e.g., Wi-Fi protocols, encryption methods)
- Wireless security threats and attacks (e.g., rogue APs, WEP/WPA cracking)
- Conducting wireless penetration testing
- Securing wireless networks and best practices

Unit 6 : Network Security: 6

- Network architecture and protocols (e.g., TCP/IP, DNS, DHCP)
- Network security threats and countermeasures
- Network penetration testing methodologies



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ep. P. K. S.

- Intrusion detection and prevention systems (IDS/IPS)

Unit 7: System Security: 6

- Operating system security fundamentals (e.g., Windows, Linux)
- Hardening operating systems and securing user accounts
- Privilege escalation and lateral movement techniques
- Endpoint security solutions and best practices

Learning Resources:

Text Books/Reference Books:

4. "The Basics of Hacking and Penetration Testing: Ethical Hacking and Penetration Testing Made Easy" by Patrick Engebretson.
5. "Hacking: The Art of Exploitation" by Jon Erickson.
6. "Penetration Testing: A Hands-On Introduction to Hacking" by Georgia Weidman.

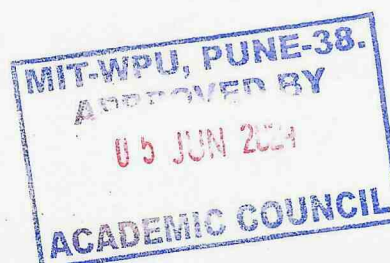
Supplementary Reading:

1. "Metasploit: The Penetration Tester's Guide" by David Kennedy, Jim O'Gorman, Devon Kearns, and Mati Aharoni.
2. "The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws" by Dafydd Stuttard and Marcus Pinto.
3. "Black Hat Python: Python Programming for Hackers and Pentesters" by Justin Seitz.

Web Resources :

Weblinks:

1. OWASP (Open Web Application Security Project): Provides resources, tools, and guides for web application security testing. <https://owasp.org/>
2. PortSwigger Web Security Academy: Offers free web security training materials and interactive labs. <https://portswigger.net/web-security>
3. Exploit Database: A repository of exploits and vulnerable software for learning and research purposes. <https://www.exploit-db.com/>



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Courses by NPTEL and MIT Open Courseware etc

Pedagogy:

- Participative Learning,
- discussions,
- problem solving,
- assignments,



Handwritten signatures and initials:
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A black signature, possibly "R. K. Kulkarni", is written to the right of the stamp.